# On the Theory of Non-Commutative Optimal Transport with Applications in Quantitative Risk Modelling

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**Abstract**

This paper presents an introduction to optimal transport theory and its extension to the non-commutative setting, with an extension to applications in actuarial science and risk modelling. The necessary mathematical foundations are developed through measure theory, probability, functional analysis, and operator algebras, culminating in the classical Monge and Kantorovich formulations of optimal transport and the Wasserstein metrics. Building on these foundations, the theory is extended to states on ($C^*$)-algebras, quantum channels, non-commutative couplings, and quantum Wasserstein distances, highlighting the relationship between classical and quantum transport frameworks. The paper concludes by examining how optimal transport provides a unifying framework for comparing probability distributions, studying robustness of risk measures, and related problems in actuarial science.

# Contents

# 1 Introduction

Optimal transport originates with Gaspard Monge, who in 1781 asked for the most economical way to move a pile of earth into a target embankment. Formalised much later, Monge's problem seeks a map $T$ pushing a source measure $\mu$ onto a target measure $\nu$ while minimising a total transport cost $\int c(x, T(x))\, d\mu(x)$. The problem is notoriously rigid, because a transport map need not exist, since mass concentrated at a single point cannot be split, and the constraint $\nu = T_*\mu$ is nonlinear. In 1942, Leonid Kantorovich relaxed the search over maps to a search over couplings, that is, joint measures $\pi$ with marginals $\mu$ and $\nu$, turning the problem into an infinite-dimensional linear program with an elegant dual. This relaxation, for which Kantorovich later shared the 1975 Nobel Memorial Prize in Economics, is the foundation of the modern theory. From the 1980s onward, work of Brenier, McCann, Otto and Villani revealed deep links to convex analysis, partial differential equations and Riemannian geometry, and established the Wasserstein distances $W_p$ as central tools in probability and analysis (26, 28, 29).

The subject of this paper is the non-commutative generalisation of this circle of ideas. In the classical picture, a probability measure $\mu$ on a space $X$ is the same data as a state, that is, a positive normalised linear functional on the commutative algebra of functions on $X$. The integral $f \mapsto \int f\, d\mu$ is precisely that state. Replacing this commutative algebra by a general $C^*$- or $W^*$-algebra $\mathcal{A}$, for instance the matrix algebra $M_n(\mathbb{C})$ whose states are the density matrices of quantum mechanics, yields quantum probability. A coupling of two states $\varphi$ on $\mathcal{A}$ and $\psi$ on $\mathcal{B}$ then becomes a state $\omega$ on the tensor product $\mathcal{A} \otimes \mathcal{B}$ whose marginals are $\varphi$ and $\psi$, exactly mirroring the classical requirement that a transport plan have marginals $\mu$ and $\nu$. Minimising an appropriate cost over such quantum couplings produces non-commutative analogues of the Kantorovich problem and of the Wasserstein distances (7, 9). The programme is motivated both from within mathematics, through non-commutative geometry and operator algebras, and by quantum information theory (20, 30), where distances between quantum states that respect an underlying metric structure are of direct physical interest.

Connections to classical probability, stochastic processes and, through Wasserstein distributionally robust optimisation and risk measures, to actuarial mathematics are indicated where they arise (14, 18).

# 2 Measure Theory and Probability

This section develops the measure theory, integration theory and probability that the rest of the paper rests upon. Standard references are Folland (13) and Bogachev (5) for measure and integration, and Billingsley (3) and Williams (31) for probability.

## 2.1 Measure Spaces and $\sigma$-Algebras

We begin by defining $\sigma$-algebras together with measures. This is then extended to probability theory.

**Definition 2.1** ($\sigma$-algebra)**.** *Let $\Omega$ be a nonempty set, and let $\mathcal{F}$ be a family of subsets of $\Omega$. Then $\mathcal{F}$ is called a $\sigma$-algebra if*

1. *$\emptyset \in \mathcal{F}$,*
2. *If $A \in \mathcal{F}$ then $A^C \in \mathcal{F}$,*

*3. If $A_n \in \mathcal{F}$ for every $n \in \mathbb{N}$, then $\bigcup_{n=1}^{\infty} A_n \in \mathcal{F}$.*

By axioms 1 and 2 we also have $\Omega = \emptyset^C \in \mathcal{F}$. Some authors take $\Omega \in \mathcal{F}$ as the first axiom instead, the two conventions being equivalent in the presence of closure under complementation. Intuitively, $\Omega$ is the set of all possible outcomes of an experiment, i.e. the sample space, and a $\sigma$-algebra is the collection of those events to which we are able to assign a size or a probability.

**Example 2.1** (A single coin toss)**.** Let $\Omega = \{H, T\}$ be the outcomes of one coin toss. The family $\mathcal{F} = \{\emptyset, \{H\}, \{T\}, \{H, T\}\}$ is a $\sigma$-algebra: it contains $\emptyset$, it is closed under complements because the complement of $\{H\}$ is $\{T\}$), and it is closed under unions. It is in fact the power set of $\Omega$, the largest possible $\sigma$-algebra on two outcomes.

**Example 2.2** (Coarser information)**.** For a die with $\Omega = \{1, 2, 3, 4, 5, 6\}$, suppose we can only observe whether the outcome is "at most 3" or "greater than 3". The events we can resolve form the $\sigma$-algebra $\mathcal{F} = \{\emptyset, \{1, 2, 3\}, \{4, 5, 6\}, \Omega\}$ generated by the single event $\{1, 2, 3\}$. This illustrates that a $\sigma$-algebra encodes exactly how much information about the outcome is available.

**Proposition 2.1.** *If $\mathcal{F}$ is a $\sigma$-algebra and $A_n \in \mathcal{F}$ for every $n \in \mathbb{N}$, then $\bigcap_{n=1}^{\infty} A_n \in \mathcal{F}$.*

*Proof.* Let $\mathcal{F}$ be a $\sigma$-algebra and let $A_1, A_2, \ldots$ be a sequence of sets in $\mathcal{F}$. Since $\mathcal{F}$ is a $\sigma$-algebra, $A_1^C, A_2^C, \cdots \in \mathcal{F}$. Furthermore $\bigcup_{n=1}^{\infty} A_n^C \in \mathcal{F}$, from which it follows that $\left(\bigcup_{n=1}^{\infty} A_n^C\right)^C \in \mathcal{F}$. Applying De Morgan's law gives $\left(\bigcup_{n=1}^{\infty} A_n^C\right)^C = \bigcap_{n=1}^{\infty} A_n \in \mathcal{F}$ as required. $\square$

**Example 2.3** (Closure under intersection)**.** On the die space with $\mathcal{F}$ the power set, take $A_1 = \{1, 2, 3, 4\}$ and $A_2 = \{3, 4, 5, 6\}$. Both lie in $\mathcal{F}$, and the proposition guarantees $A_1 \cap A_2 = \{3, 4\} \in \mathcal{F}$, which is indeed a subset of $\Omega$ and hence a member of the power set.

**Example 2.4.** Let $\Omega$ be any non-empty set. Then the set $\mathcal{F}_s = \{\emptyset, \Omega\}$ is a $\sigma$-algebra of subsets of $\Omega$. Moreover, $\mathcal{F}_s$ is the smallest $\sigma$-algebra of $\Omega$, called the trivial $\sigma$-algebra. The largest $\sigma$-algebra of $\Omega$ is the power set $\mathcal{F}_l = \mathcal{P}[\Omega]$.

**Theorem 2.1** (Properties of $\sigma$-Algebras)**.** *Let $\mathcal{F}$ be a $\sigma$-algebra of subsets of $\Omega$. Then $\mathcal{F}$ has the following properties:*

*(S1) If $A, B \in \mathcal{F}$ then $A \cup B \in \mathcal{F}$.*

*(S2) If $A, B \in \mathcal{F}$ then $A \cap B \in \mathcal{F}$.*

*(S3) If $A, B \in \mathcal{F}$ then $A \setminus B \in \mathcal{F}$.*

We give a proof of S3 only and assume the rest.

*Proof.* Let $\Omega$ be a non-empty set with $\mathcal{F}$ a $\sigma$-algebra of subsets of $\Omega$. Assume that $A$ and $B$ are two sets in $\mathcal{F}$. Since $A, B \in \mathcal{F}$, it follows from the definition of a $\sigma$-algebra that $A^C, B^C \in \mathcal{F}$. We apply De Morgan's law to $A \setminus B = A \cap B^C$ to obtain $A \cap B^C = (A^C \cup B)^C$. From S1 we have that $A^C \cup B \in \mathcal{F}$. It therefore follows that $A \setminus B = (A^C \cup B)^C \in \mathcal{F}$ as required. $\square$

**Example 2.5** (Set difference of events)**.** On $\Omega = \{1, 2, 3, 4, 5, 6\}$ with $\mathcal{F}$ the power set, let $A = \{1, 2, 3, 4\}$, i.e. the event of "obtaining at most 4" and $B = \{2, 4, 6\}$, i.e. the event of "obtaining only even numbers". Then $A \setminus B = \{1, 3\}$, the event "at most 4 and odd", which by S3 is again a member of $\mathcal{F}$.

We now formulate Borel sets. Consider a non-empty set $\Omega$ and let $\mathcal{B}$ be any family of $\sigma$-algebras of subsets of $\Omega$. We assume that the intersection $\bigcap\limits_{\Sigma \in \mathcal{B}} \Sigma$ of all $\sigma$-algebras in $\mathcal{B}$ is again a $\sigma$-algebra of subsets of $\Omega$. Now let $\mathcal{A}$ be any family of subsets of $\Omega$ and define $\mathcal{B} = \{\Sigma : \Sigma \text{ is a } \sigma\text{-algebra of subsets of } \Omega, \mathcal{A} \subseteq \Sigma\}$. Then the $\sigma$-algebra generated by $\mathcal{A}$ is $\sigma(\mathcal{A}) = \bigcap\limits_{\Sigma \in \mathcal{B}} \Sigma$, the smallest $\sigma$-algebra of subsets of $\Omega$ that contains $\mathcal{A}$.

**Example 2.6.** For any non-empty set $\Omega$, the $\sigma$-algebra generated by $\emptyset$ is $\sigma(\emptyset) = \{\emptyset, \Omega\}$, the trivial $\sigma$-algebra, because $\{\emptyset, \Omega\}$ is the smallest $\sigma$-algebra containing the empty set.

**Example 2.7** (Generating from one event)**.** On $\Omega = \{1, 2, 3, 4, 5, 6\}$ the family $\mathcal{A} = \{\{1, 2, 3\}\}$ generates $\sigma(\mathcal{A}) = \{\emptyset, \{1, 2, 3\}, \{4, 5, 6\}, \Omega\}$, since any $\sigma$-algebra containing $\{1, 2, 3\}$ must also contain its complement and the whole space.

**Definition 2.2** (Borel $\sigma$-Algebra)**.** *Let $(\Omega, \tau)$ be a topological space, where $\tau$ denotes the collection of open subsets of $\Omega$. The smallest $\sigma$-algebra containing every open set is called the Borel $\sigma$-algebra over $\Omega$, written $\mathcal{B}[\Omega] := \sigma(\tau)$.*

A Borel subset of a topological space $\Omega$ is an element of the Borel $\sigma$-algebra over $\Omega$.

**Example 2.8** (Borel sets of the real line)**.** On $\mathbb{R}$ with its usual topology, $\mathcal{B}[\mathbb{R}]$ is generated by the open intervals $(a, b)$. It therefore contains every open set, every closed set (as complements), every half-line $(-\infty, a]$, every singleton $\{a\} = \bigcap_n (a - \frac{1}{n}, a + \frac{1}{n})$, and every countable set such as $\mathbb{Q}$. This is the $\sigma$-algebra on which the Lebesgue measure is built below.

**Definition 2.3** (Measure Space)**.** *Let $\Omega$ be any set with $\mathcal{F}$ a $\sigma$-algebra of subsets of $\Omega$. Define $\mu : \mathcal{F} \to [0, \infty]$ as a function such that*

1. $\mu(\emptyset) = 0$,
2. *If $\{A_n\}_{n \in \mathbb{N}}$ is a sequence of pairwise disjoint sets in $\mathcal{F}$ then $\mu\left(\bigcup\limits_{n \in \mathbb{N}} A_n\right) = \sum\limits_{n=1}^{\infty} \mu(A_n)$ (countable additivity).*

*Then the triple $(\Omega, \mathcal{F}, \mu)$ is called a measure space, and members of $\mathcal{F}$ are called measurable sets.*

**Example 2.9** (Counting measure)**.** Let $\Omega$ be any set with $\mathcal{F} = \mathcal{P}[\Omega]$, and set $\mu(A) = \#A$, the number of elements of $A$ (with $\mu(A) = \infty$ if $A$ is infinite). Then $\mu(\emptyset) = 0$ and disjoint sets have sizes that add, so $\mu$ is a measure, called the counting measure. For $\Omega = \{1, 2, 3, 4, 5, 6\}$ we have $\mu(\{2, 4, 6\}) = 3$.

**Example 2.10** (Dirac measure)**.** Fix a point $x_0 \in \Omega$ and define $\delta_{x_0}(A) = 1$ if $x_0 \in A$ and 0 otherwise. This is a measure that places all of its mass at $x_0$. It is the mathematical model of a deterministic outcome, and it reappears below as the simplest probability distribution.

**Theorem 2.2** (Properties of Measure Spaces)**.** *Let $(\Omega, \mathcal{F}, \mu)$ be a measure space, then $(\Omega, \mathcal{F}, \mu)$ has the following properties:*

*(M1)* If $A, B \in \mathcal{F}$ and $A \cap B = \emptyset$ then $\mu(A \cup B) = \mu(A) + \mu(B)$.

*(M2)* If $A, B \in \mathcal{F}$ and $A \subseteq B$ then $\mu(A) \leq \mu(B)$.

*(M3)* $\mu(A \cup B) \leq \mu(A) + \mu(B)$ for all $A, B \in \mathcal{F}$.

*(M4)* If $A_n \in \mathcal{F}$ for every $n \in \mathbb{N}$ then $\mu(\bigcup_{n\in\mathbb{N}} A_n) \leq \sum_{n=1}^{\infty} \mu(A_n)$ (countable subadditivity).

*(M5)* If $A_n \in \mathcal{F}$ for every $n$ and $A_n \subseteq A_{n+1}$ for every $n$ then $\mu(\bigcup_{n\in\mathbb{N}} A_n) = \lim\limits_{n\to\infty} \mu(A_n) = \sup\limits_{n\in\mathbb{N}} \mu(A_n)$ (continuity from below).

*(M6)* If $A_n \in \mathcal{F}$ for every $n$, $A_{n+1} \subseteq A_n$ for every $n$, and $\mu(A_{n_0}) < \infty$ for some $n_0 \in \mathbb{N}$, then $\mu(\bigcap_{n\in\mathbb{N}} A_n) = \lim\limits_{n\to\infty} \mu(A_n) = \inf\limits_{n\in\mathbb{N}} \mu(A_n)$ (continuity from above).

We give a proof of M3 only and assume the rest.

*Proof.* Let $(\Omega, \mathcal{F}, \mu)$ be a measure space with $A, B \in \mathcal{F}$. It follows from M1 that $\mu(A \cup B) = \mu(A \cup (B \setminus A)) = \mu(A) + \mu(B \setminus A)$ because $A$ and $B \setminus A$ are disjoint. Moreover $B \setminus A \subseteq B$, so by M2 we have $\mu(B \setminus A) \leq \mu(B)$. Therefore $\mu(A \cup B) = \mu(A) + \mu(B \setminus A) \leq \mu(A) + \mu(B)$ as required. $\square$

**Example 2.11** (Continuity from below)**.** Take Lebesgue measure $\lambda$ on $\mathbb{R}$ (constructed below) and the increasing sets $A_n = [0, 1 - \frac{1}{n}]$. Then $\bigcup_n A_n = [0, 1)$ and $\lambda(A_n) = 1 - \frac{1}{n} \to 1 = \lambda([0,1))$, illustrating property M5: the measure of an increasing union is the limit of the measures.

## 2.2 Outer Measures and the Lebesgue Measure

We now introduce an elementary method of constructing measures via Carathéodory's method, together with the Lebesgue measure.

**Definition 2.4** (Outer Measure)**.** *Let $\Omega$ be any set. Define a function $\theta : \mathcal{P}[\Omega] \to [0, \infty]$ such that*

1. $\theta(\emptyset) = 0$,
2. *If $A \subseteq B \subseteq \Omega$ then $\theta(A) \leq \theta(B)$ (monotonicity),*
3. *For every sequence $\{A_n\}_{n \in \mathbb{N}}$ of subsets of $\Omega$, $\theta(\bigcup_{n\in\mathbb{N}} A_n) \leq \sum_{n=1}^{\infty} \theta(A_n)$ (countable subadditivity).*

*Then $\theta$ is called an outer measure on $\Omega$.*

Intuitively, an outer measure lets us assign a size to *every* subset by covering it with sets of known size and taking the most efficient cover. Suppose we wish to measure a set $\Omega$. We cover it with sets $A_n$ whose measures are known and sum those measures. Refining the cover, we approach the infimum over all covers, which is the outer measure.

**Example 2.12** (An outer measure from a single cover)**.** On any $\Omega$, fix $x_0 \in \Omega$ and set $\theta(A) = 1$ if $x_0 \in A$ and $\theta(A) = 0$ otherwise. Then $\theta(\emptyset) = 0$, $\theta$ is monotone, and it is subadditive, so $\theta$ is an outer measure. It is defined on *all* subsets, unlike the Dirac measure, which we regarded as defined on a $\sigma$-algebra.

**Theorem 2.3** (Carathéodory's Method)**.** *Let $\Omega$ be any set and $\theta$ an outer measure on $\Omega$. Let*

$$\mathcal{F} = \left\{ A : A \subseteq \Omega \wedge \theta(B) = \theta(B \cap A) + \theta(B \setminus A) \textit{ for every } B \subseteq \Omega \right\}. \tag{1}$$

*Then $\mathcal{F}$ is a $\sigma$-algebra of subsets of $\Omega$. Moreover, define $\mu : \mathcal{F} \to [0, \infty]$ by $\mu(A) = \theta(A)$ for $A \in \mathcal{F}$. Then $(\Omega, \mathcal{F}, \mu)$ is a measure space.*

**Example 2.13** (The Carathéodory criterion in action)**.** For the outer measure of the previous example, a set $A$ is measurable when, for every test set $B$, splitting $B$ by $A$ does not lose mass. One checks $\{x_0\}$ is measurable: if $x_0 \in B$ then $\theta(B) = 1 = \theta(B \cap \{x_0\}) + \theta(B \setminus \{x_0\}) = 1 + 0$, and if $x_0 \notin B$ both sides are 0. Thus Carathéodory's method turns a crude outer measure into a genuine measure on the sets that "cut cleanly".

**Definition 2.5** (Half-Open Intervals in $\mathbb{R}^n$)**.** *A half-open interval in $\mathbb{R}^n$ is a set*

$$[\overline{a}, \overline{b}[ = \{\overline{x} \in \mathbb{R}^n : \alpha_i \leq \xi_i < \beta_i \text{ for all } i \leq n\}, \tag{2}$$

*where $\overline{a} = (\alpha_1, \ldots, \alpha_n)$ and $\overline{b} = (\beta_1, \ldots, \beta_n) \in \mathbb{R}^n$, and $\overline{x} = (\xi_1, \ldots, \xi_n)$.*

**Example 2.14.** In $\mathbb{R}^2$, with $\overline{a} = (0, 0)$ and $\overline{b} = (1, 2)$, the half-open interval $[\overline{a}, \overline{b}[$ is the rectangle $\{(x, y) : 0 \leq x < 1,\ 0 \leq y < 2\}$, which includes its lower and left edges but not its upper and right edges.

**Definition 2.6** ($n$-Dimensional Volume)**.** *Let $I = [\overline{a}, \overline{b}[ \subseteq \mathbb{R}^n$ be a half-open interval. If $I \neq \emptyset$ then $\alpha_i = \inf\{\xi_i : \overline{x} \in I\}$ and $\beta_i = \sup\{\xi_i : \overline{x} \in I\}$ for every $i \leq n$. The $n$-dimensional volume of $I$ is*

$$\lambda I = \begin{cases} \prod_{i=1}^n (\beta_i - \alpha_i), & \text{if } \alpha_i < \beta_i \text{ for every } i, \\ 0, & \text{if } I = \emptyset. \end{cases} \tag{3}$$

**Example 2.15.** For the rectangle above, $\lambda I = (1 - 0)(2 - 0) = 2$, matching the usual area. In one dimension, $\lambda[a, b[ = b - a$ is the length of the interval.

**Definition 2.7** (Lebesgue Outer Measure on $\mathbb{R}^n$)**.** *Define a function $\theta : \mathcal{P}[\mathbb{R}^n] \to [0, \infty]$ such that*

$$\theta[A] = \inf \left\{ \sum_{i=1}^{\infty} \lambda I_i : \{I_i\}_{i \in \mathbb{N}} \text{ is a sequence of half-open intervals such that } A \subseteq \bigcup_{i \in \mathbb{N}} I_i \right\}. \tag{4}$$

*Then $\theta$ is called the Lebesgue outer measure on $\mathbb{R}^n$.*

Let $\theta$ denote the Lebesgue outer measure on $\mathbb{R}^n$. By Carathéodory's method we construct a measure from $\theta$. The sets $B$ for which $\theta[A \cap B] + \theta[A \setminus B] = \theta[A]$ for every $A \subseteq \mathbb{R}^n$ are the Lebesgue measurable sets, and the resulting measure is the Lebesgue measure on $\mathbb{R}^n$, denoted $\lambda$.

**Example 2.16** (Lebesgue measure of a point and of $\mathbb{Q}$)**.** A single point $\{a\} \subseteq \mathbb{R}$ can be covered by $[a, a + \frac{1}{n}[$ of length $\frac{1}{n}$, so $\theta[\{a\}] \leq \frac{1}{n}$ for every $n$, giving $\lambda(\{a\}) = 0$. By countable additivity, any countable set has measure zero, so $\lambda(\mathbb{Q}) = 0$. Thus the rationals, though dense, are negligible for Lebesgue measure, while $\lambda([0, 1]) = 1$.

## 2.3 Measurable Functions and the Pushforward Measure

We now discuss mappings between measurable spaces, which are used to introduce the Lebesgue integral and to formulate random variables in probability theory.

**Theorem 2.4** (Subspace $\sigma$-Algebras)**.** *Let $\Omega$ be any set and $\mathcal{F}$ a $\sigma$-algebra of subsets of $\Omega$. Let $A$ be any subset of $\Omega$ and set $\mathcal{F}_A = \{B \cap A : B \in \mathcal{F}\}$. Then $\mathcal{F}_A$ is a $\sigma$-algebra of subsets of $A$.*

We call $\mathcal{F}_A$ the subspace $\sigma$-algebra on $A$, or the trace of $\mathcal{F}$ on $A$.

*Proof.* Let $\mathcal{F}_A = \{B \cap A : B \in \mathcal{F}\}$.

i. $\mathcal{F}$ is a $\sigma$-algebra, so $\emptyset \in \mathcal{F}$, therefore $\emptyset = \emptyset \cap A \in \mathcal{F}_A$.

ii. Let $X \in \mathcal{F}_A$, then there exists $B \in \mathcal{F}$ such that $X = B \cap A$. Moreover $A \setminus X = (\Omega \setminus B) \cap A \in \mathcal{F}_A$ because $\Omega \setminus B \in \mathcal{F}$.

iii. Let $\{X_i\}_{i\in\mathbb{N}}$ be any sequence in $\mathcal{F}_A$. For every $i$ take $B_i \in \mathcal{F}$ with $X_i = B_i \cap A$. Since $\bigcup_i B_i \in \mathcal{F}$ we have $\bigcup_i X_i = (\bigcup_i B_i) \cap A \in \mathcal{F}_A$.

Therefore $\mathcal{F}_A$ is a $\sigma$-algebra of subsets of $A$. □

**Example 2.17.** Let $\Omega = \{1, 2, 3, 4\}$ with $\mathcal{F} = \{\emptyset, \{1, 2\}, \{3, 4\}, \Omega\}$, and let $A = \{1, 2, 3\}$. Then the trace is $\mathcal{F}_A = \{\emptyset, \{1, 2\}, \{3\}, \{1, 2, 3\}\}$, obtained by intersecting each member of $\mathcal{F}$ with $A$. It is a $\sigma$-algebra of subsets of $A$.

We state the following theorem without proof.

**Theorem 2.5** (Measurable Conditions)**.** *Let $\Omega$ be any set and $\mathcal{F}$ a $\sigma$-algebra of subsets of $\Omega$ with $A \subseteq \Omega$. Write the trace of $\mathcal{F}$ on $A$ as $\mathcal{F}_A$. For any function $\phi : A \to \mathbb{R}$ the following four statements are equivalent, so that if one holds then all hold:*

*i.* $\{x : \phi(x) < a\} \in \mathcal{F}_A$ *for every* $a \in \mathbb{R}$,

*ii.* $\{x : \phi(x) \le a\} \in \mathcal{F}_A$ *for every* $a \in \mathbb{R}$,

*iii.* $\{x : \phi(x) > a\} \in \mathcal{F}_A$ *for every* $a \in \mathbb{R}$,

*iv.* $\{x : \phi(x) \ge a\} \in \mathcal{F}_A$ *for every* $a \in \mathbb{R}$.

**Definition 2.8** (Measurable Function)**.** *Let $\Omega$ be any set and $\mathcal{F}$ a $\sigma$-algebra of subsets of $\Omega$ with $A \subseteq \Omega$. A function $\phi : A \to \mathbb{R}$ is called $\mathcal{F}$-measurable if it satisfies the (equivalent) measurable conditions.*

If $\Omega = \mathbb{R}^n$ and $\mathcal{F}$ is its Borel $\sigma$-algebra then we say $\phi$ is Borel measurable. If $\mathcal{F}$ is the $\sigma$-algebra of Lebesgue measurable sets then $\phi$ is Lebesgue measurable.

**Example 2.18** (Continuous functions are measurable)**.** Every continuous $\phi : \mathbb{R} \to \mathbb{R}$ is Borel measurable, because $\{x : \phi(x) < a\} = \phi^{-1}((-\infty, a))$ is open, hence Borel, for each $a$. For instance $\phi(x) = x^2$ is measurable, with $\{x : x^2 < a\} = (-\sqrt{a}, \sqrt{a})$ for $a > 0$.

**Definition 2.9** (Pushforward Measure)**.** *Let $(\Omega_1, \mathcal{F}_1)$ and $(\Omega_2, \mathcal{F}_2)$ be measurable spaces, let $\phi : \Omega_1 \to \Omega_2$ be a measurable function, and let $\mu : \mathcal{F}_1 \to [0, +\infty]$ be a measure. The pushforward of $\mu$ by $\phi$ is the measure $\phi_*(\mu) : \mathcal{F}_2 \to [0, +\infty]$ defined by*

$$\phi_*(\mu)(B) = \mu(\phi^{-1}(B)) \qquad \textit{for all } B \in \mathcal{F}_2. \tag{5}$$

*Measurability of $\phi$ guarantees $\phi^{-1}(B) \in \mathcal{F}_1$, so the right-hand side is well defined.*

The pushforward transfers a measure from one space to another along a map. In probability, the distribution of a random variable is the pushforward of the probability measure by that variable.

**Example 2.19** (Pushforward by a scaling map)**.** Let $\mu$ be Lebesgue measure on $[0, 1]$ and $\phi(x) = 2x$, so $\phi : [0, 1] \to [0, 2]$. For an interval $[c, d] \subseteq [0, 2]$ we have $\phi^{-1}([c, d]) = [\frac{c}{2}, \frac{d}{2}]$, hence $\phi_*(\mu)([c, d]) = \mu([\frac{c}{2}, \frac{d}{2}]) = \frac{d-c}{2}$. The pushforward is thus one half of Lebesgue measure on $[0, 2]$: stretching the line by a factor of two halves the density.

**Definition 2.10** (Indicator Function)**.** *Let $(\Omega, \mathcal{F}, \mu)$ be a measure space. For any measurable set $A \in \mathcal{F}$, the indicator function $\mathbf{1}_A : \Omega \to \{0,1\}$ of $A$ is*

$$\mathbf{1}_A(x) = \begin{cases} 1, & \text{if } x \in A, \\ 0, & \text{if } x \in \Omega \setminus A. \end{cases} \tag{6}$$

*Since $A \in \mathcal{F}$, the function $\mathbf{1}_A$ is $\mathcal{F}$-measurable. We reserve the term characteristic function for the Fourier transform $t \mapsto \mathbb{E}[e^{itX}]$ of a random variable, and use indicator function here to avoid a clash of terminology.*

**Example 2.20.** On $\mathbb{R}$, the indicator of $A = [0,1]$ is the function equal to 1 on $[0,1]$ and 0 elsewhere, a single unit step. Its Lebesgue integral, defined next, is the area under it, namely 1.

**Definition 2.11** (Simple Function)**.** *Let $(\Omega, \mathcal{F}, \mu)$ be a measure space. A simple function on $\Omega$ is a function of the form $\sum_{i=1}^{n} a_i \mathbf{1}_{X_i}$, where the $X_i$ are measurable sets of finite measure and $a_i \in \mathbb{R}$ for $i = 1, 2, \ldots, n$.*

**Example 2.21** (A step function)**.** On $\mathbb{R}$, the function $f = 2 \cdot \mathbf{1}_{[0,1)} + 5 \cdot \mathbf{1}_{[1,3)}$ is simple: it takes the value 2 on $[0,1)$, the value 5 on $[1,3)$, and 0 elsewhere. Simple functions are the finite-valued building blocks from which the Lebesgue integral is constructed.

We state the following theorem about simple functions without proof, after which we define their integral.

**Theorem 2.6.** *Let $(\Omega, \mathcal{F}, \mu)$ be a measure space.*

1. *Every simple function on $\Omega$ is measurable.*
2. *If $f, g : \Omega \to \mathbb{R}$ are simple functions, then so is $f + g$.*
3. *If $f : \Omega \to \mathbb{R}$ is a simple function and $c \in \mathbb{R}$, then $cf : \Omega \to \mathbb{R}$ is a simple function.*
4. *The constant zero function is simple.*

**Example 2.22.** With $f = 2 \cdot \mathbf{1}_{[0,1)}$ and $g = 3 \cdot \mathbf{1}_{[0,2)}$, the sum $f + g = 5 \cdot \mathbf{1}_{[0,1)} + 3 \cdot \mathbf{1}_{[1,2)}$ is again simple, and $2f = 4 \cdot \mathbf{1}_{[0,1)}$ is simple, illustrating parts 2 and 3.

**Definition 2.12** (Integration of Simple Functions)**.** *Let $(\Omega, \mathcal{F}, \mu)$ be a measure space and let $f = \sum_{i=1}^{n} a_i \mathbf{1}_{X_i}$ be a non-negative simple function written in canonical form, that is, with the $X_i \in \mathcal{F}$ pairwise disjoint and the $a_i \geq 0$ distinct. Its integral is*

$$\int_\Omega f \, d\mu = \sum_{i=1}^{n} a_i \, \mu(X_i). \tag{7}$$

*This value is independent of the representation of $f$, so the integral is well defined on all simple functions.*

**Example 2.23.** For $f = 2\cdot\mathbf{1}_{[0,1)} + 5\cdot\mathbf{1}_{[1,3)}$ against Lebesgue measure, $\int_\mathbb{R} f \, d\lambda = 2\cdot\lambda([0,1)) + 5\cdot\lambda([1,3)) = 2 \cdot 1 + 5 \cdot 2 = 12$, which is the total area of the two rectangles under the step function.

**Definition 2.13** (Lebesgue Integral)**.** *Let $(\Omega, \mathcal{F}, \mu)$ be a measure space and let $f : \Omega \to \mathbb{R}$ be measurable. Write $f = f^+ - f^-$ where $f^+ = \max(f, 0)$ and $f^- = \max(-f, 0)$. If $\int f^+ d\mu < \infty$ and $\int f^- d\mu < \infty$, then $f$ is called Lebesgue integrable and its integral is*

$$\int_\Omega f \, d\mu = \int_\Omega f^+ d\mu - \int_\Omega f^- d\mu. \tag{8}$$

*Here $\int f^+ d\mu$ and $\int f^- d\mu$ are defined as suprema of the integrals of simple functions below $f^+$ and $f^-$ respectively.*

**Example 2.24** (Integrating a function with sign)**.** Let $f = \mathbf{1}_{[0,1)} - 3 \cdot \mathbf{1}_{[1,2)}$ on $\mathbb{R}$. Then $f^+ = \mathbf{1}_{[0,1)}$ and $f^- = 3 \cdot \mathbf{1}_{[1,2)}$, so $\int f \, d\lambda = \int f^+ d\lambda - \int f^- d\lambda = 1 - 3 = -2$. Splitting into positive and negative parts is exactly how the Lebesgue integral handles cancellation.

The following two convergence theorems are the workhorses of Lebesgue integration and are used implicitly whenever integrals and limits are interchanged. We state them without proof.

**Theorem 2.7** (Monotone Convergence Theorem)**.** *Let $(\Omega, \mathcal{F}, \mu)$ be a measure space and let $f_n : \Omega \to [0, \infty]$ be measurable with $f_n \leq f_{n+1}$ for every $n$ and $f_n \to f$ pointwise. Then*

$$\int_\Omega f \, d\mu = \lim_{n \to \infty} \int_\Omega f_n \, d\mu. \tag{9}$$

**Example 2.25.** On $[0, 1]$ let $f_n = (1 - \frac{1}{n})\mathbf{1}_{[0,1]}$, an increasing sequence with limit $f = \mathbf{1}_{[0,1]}$. Then $\int f_n \, d\lambda = 1 - \frac{1}{n}$ increases to $1 = \int f \, d\lambda$, as the theorem predicts.

**Theorem 2.8** (Dominated Convergence Theorem)**.** *Let $f_n : \Omega \to \mathbb{R}$ be measurable with $f_n \to f$ pointwise, and suppose there is an integrable $g \geq 0$ with $|f_n| \leq g$ for all $n$. Then $f$ is integrable and*

$$\int_\Omega f \, d\mu = \lim_{n \to \infty} \int_\Omega f_n \, d\mu. \tag{10}$$

**Example 2.26.** On $[0, 1]$ let $f_n(x) = \dfrac{x}{1 + nx^2}$. Then $f_n \to 0$ pointwise, and $|f_n| \leq g$ with $g(x) = \frac{1}{2}$ (since $\frac{x}{1+nx^2} \leq \frac{1}{2\sqrt{n}} \leq \frac{1}{2}$), which is integrable on $[0, 1]$. The theorem gives $\int_0^1 f_n \, d\lambda \to 0$, letting us pass the limit inside the integral even though no monotonicity is present.

## 2.4 The Product Measure

**Definition 2.14** (Product $\sigma$-Algebras)**.** *Let $(\Omega_1, \mathcal{F}_1)$, $(\Omega_2, \mathcal{F}_2)$ be measurable spaces. The product $\sigma$-algebra $\mathcal{F}_1 \otimes \mathcal{F}_2$ is the $\sigma$-algebra on $\Omega_1 \times \Omega_2$ generated by the measurable rectangles:*

$$\mathcal{F}_1 \otimes \mathcal{F}_2 = \sigma\big(\{A \times B : A \in \mathcal{F}_1,\ B \in \mathcal{F}_2\}\big). \tag{11}$$

*The product of $(\Omega_1, \mathcal{F}_1)$ and $(\Omega_2, \mathcal{F}_2)$ is the measurable space $(\Omega_1 \times \Omega_2, \mathcal{F}_1 \otimes \mathcal{F}_2)$.*

**Example 2.27.** Taking $\Omega_1 = \Omega_2 = \mathbb{R}$ with the Borel $\sigma$-algebra, the product $\mathcal{B}[\mathbb{R}] \otimes \mathcal{B}[\mathbb{R}]$ is generated by rectangles $(a, b) \times (c, d)$ and equals the Borel $\sigma$-algebra $\mathcal{B}[\mathbb{R}^2]$ of the plane. Thus the natural $\sigma$-algebra on $\mathbb{R}^2$ is the product of the ones on each axis.

**Definition 2.15** (Premeasure)**.** *Let $\mathcal{E}$ be an algebra of subsets of a set $\Omega$. A function $\phi : \mathcal{E} \to [0, \infty]$ is called a premeasure on $\mathcal{E}$ if:*

1. $\phi(\emptyset) = 0$,
2. If $\{A_i\}_{i\in\mathbb{N}}$ is a sequence of pairwise disjoint sets in $\mathcal{E}$ with $\bigcup_{i=1}^{\infty} A_i \in \mathcal{E}$, then $\phi(\bigcup_{i=1}^{\infty} A_i) = \sum_{i=1}^{\infty} \phi(A_i)$.

A premeasure is like a measure but is only required to be defined on an algebra (closed under finite operations) rather than a full $\sigma$-algebra. It is the starting data that Carathéodory's method extends to a genuine measure.

**Example 2.28.** Let $\mathcal{E}$ be the algebra of finite unions of half-open intervals in $\mathbb{R}$, and let $\phi([a,b[) = b-a$ be their total length. Then $\phi$ is a premeasure, and Carathéodory's method extends it to the Lebesgue measure on the Borel $\sigma$-algebra.

**Definition 2.16** (Outer Measure Induced by a Premeasure)**.** *Let $\mathcal{E}$ be an algebra of subsets of $\Omega$ and $\phi : \mathcal{E} \to [0,\infty]$ a premeasure. The outer measure $\phi^* : \mathcal{P}(\Omega) \to [0,\infty]$ associated with $\phi$ is defined, for $B \subseteq \Omega$, by*

$$\phi^*(B) = \inf\left\{ \sum_{i=1}^{\infty} \phi(A_i) : B \subseteq \bigcup_{i=1}^{\infty} A_i,\ A_i \in \mathcal{E} \right\}. \tag{12}$$

**Example 2.29.** For the length premeasure on $\mathbb{R}$, the induced outer measure $\phi^*$ is exactly the Lebesgue outer measure. Covering $\{0\}$ by $[0, \frac{1}{n}[$ shows $\phi^*(\{0\}) = 0$, recovering that points have length zero.

**Definition 2.17** (Product Measure)**.** *Let $(A, \mathcal{F}_1, \mu)$ and $(B, \mathcal{F}_2, \nu)$ be $\sigma$-finite measure spaces. The product measure $\mu \otimes \nu$ on the product $\sigma$-algebra $\mathcal{F}_1 \otimes \mathcal{F}_2$ is the unique measure satisfying*

$$(\mu \otimes \nu)(E \times F) = \mu(E)\,\nu(F) \tag{13}$$

*for all $E \in \mathcal{F}_1$, $F \in \mathcal{F}_2$.*

The $\sigma$-finiteness hypothesis is what guarantees uniqueness of the product measure, without it several distinct measures may agree on rectangles. The following identity, Tonelli's theorem, holds for every non-negative measurable $f : A \times B \to [0,\infty]$:

$$\int_{A\times B} f(a,b)\, d(\mu\otimes\nu)(a,b) = \int_A \int_B f(a,b)\, d\nu(b)\, d\mu(a) = \int_B \int_A f(a,b)\, d\mu(a)\, d\nu(b). \tag{14}$$

**Example 2.30** (Area as a product measure)**.** Let $\mu = \nu = \lambda$ be Lebesgue measure on $\mathbb{R}$. Then $\lambda \otimes \lambda$ is the area (two-dimensional Lebesgue) measure on $\mathbb{R}^2$, and for the rectangle $E \times F = [0,2] \times [0,3]$ we get $(\lambda \otimes \lambda)(E \times F) = \lambda([0,2])\,\lambda([0,3]) = 2 \cdot 3 = 6$, the usual area.

**Definition 2.18** (Product Premeasure)**.** *Let $(\Omega_1, \mathcal{F}_1, \mu)$ and $(\Omega_2, \mathcal{F}_2, \nu)$ be measure spaces, and let $\mathcal{E}$ be the algebra of finite disjoint unions of measurable rectangles in $\Omega_1 \times \Omega_2$. The product premeasure $\phi : \mathcal{E} \to [0,\infty]$ assigns to a single measurable rectangle $A \times B$ (with $A \in \mathcal{F}_1$, $B \in \mathcal{F}_2$) the value $\phi(A \times B) = \mu(A)\,\nu(B)$, and is extended additively to finite disjoint unions of rectangles.*

**Example 2.31.** For $\mu = \nu = \lambda$, the L-shaped region $([0,2] \times [0,1]) \cup ([0,1] \times [1,2])$ is a disjoint union of two rectangles, and the product premeasure assigns it $2 \cdot 1 + 1 \cdot 1 = 3$, its area.

**Definition 2.19** (Product Outer Measure)**.** *Let* $(\Omega_1, \mathcal{F}_1, \mu)$ *and* $(\Omega_2, \mathcal{F}_2, \nu)$ *be measure spaces. The product outer measure* $(\mu \otimes \nu)^* : \mathcal{P}(\Omega_1 \times \Omega_2) \to [0, \infty]$ *is defined, for* $X \subseteq \Omega_1 \times \Omega_2$*, by*

$$(\mu \otimes \nu)^*(X) = \inf \left\{ \sum_{i=1}^{\infty} \mu(A_i)\, \nu(B_i) : X \subseteq \bigcup_{i=1}^{\infty} (A_i \times B_i),\ A_i \in \mathcal{F}_1,\ B_i \in \mathcal{F}_2 \right\}. \tag{15}$$

*Restricting* $(\mu \otimes \nu)^*$ *to the Carathéodory-measurable sets recovers the product measure* $\mu \otimes \nu$*.*

**Example 2.32.** The diagonal $D = \{(x, x) : x \in [0, 1]\} \subseteq \mathbb{R}^2$ can be covered by $n$ squares of side $\frac{1}{n}$ along the line $y = x$, of total area $n \cdot \frac{1}{n^2} = \frac{1}{n} \to 0$. Hence $(\lambda \otimes \lambda)^*(D) = 0$: a line segment has zero area, as expected.

**Theorem 2.9** (Fubini's Theorem)**.** *Let* $(\Omega_1, \mathcal{F}_1, \mu)$ *and* $(\Omega_2, \mathcal{F}_2, \nu)$ *be* $\sigma$*-finite measure spaces, and let* $\phi : \Omega_1 \times \Omega_2 \to \mathbb{C}$ *be* $(\mathcal{F}_1 \otimes \mathcal{F}_2)$*-measurable. If* $\phi$ *is integrable, that is*

$$\int_{\Omega_1 \times \Omega_2} |\phi|\ d(\mu \otimes \nu) = \int_{\Omega_2} \left( \int_{\Omega_1} |\phi(x, y)|\, d\mu(x) \right) d\nu(y) < \infty, \tag{16}$$

*the last equality holding by Tonelli's theorem since* $|\phi| \geq 0$*, then the two iterated integrals of* $\phi$ *agree with its integral over the product:*

$$\int_{\Omega_1 \times \Omega_2} \phi\ d(\mu \otimes \nu) = \int_{\Omega_2} \left( \int_{\Omega_1} \phi(x, y)\, d\mu(x) \right) d\nu(y) = \int_{\Omega_1} \left( \int_{\Omega_2} \phi(x, y)\, d\nu(y) \right) d\mu(x). \tag{17}$$

**Example 2.33** (Evaluating a double integral either way)**.** Let $\phi(x, y) = xy$ on $[0, 1] \times [0, 1]$ against $\lambda \otimes \lambda$. Since $\phi \geq 0$ and $\int |\phi| < \infty$, Fubini applies:

$$\int_0^1 \int_0^1 xy\, dx\, dy = \int_0^1 y \Big( \int_0^1 x\, dx \Big) dy = \int_0^1 y \cdot \tfrac{1}{2}\, dy = \tfrac{1}{4}, \tag{18}$$

and integrating in the other order gives the same value $\frac{1}{4}$. Fubini's theorem is what justifies computing a double integral as an iterated one.

## 2.5 Probability Measures

**Definition 2.20** (Probability Measure)**.** *Let* $\Omega$ *be a nonempty set, and let* $\mathcal{F}$ *be a* $\sigma$*-algebra of subsets of* $\Omega$*. A function* $\mathbb{P} : \mathcal{F} \to [0, 1]$ *is called a probability measure if*

1. $\mathbb{P}(\emptyset) = 0$,
2. $\mathbb{P}(\Omega) = 1$,
3. *If* $\{A_n\}_{n \in \mathbb{N}}$ *is a sequence of pairwise disjoint sets in* $\mathcal{F}$*, then* $\mathbb{P}\big( \bigcup_{n=1}^{\infty} A_n \big) = \sum_{n=1}^{\infty} \mathbb{P}(A_n)$.

*The triple* $(\Omega, \mathcal{F}, \mathbb{P})$ *is called a probability space. A probability measure is precisely a measure with the additional normalisation* $\mathbb{P}(\Omega) = 1$*.*

**Example 2.34** (A fair die)**.** Let $\Omega = \{1, 2, 3, 4, 5, 6\}$ with $\mathcal{F}$ the power set, and set $\mathbb{P}(A) = \frac{\#A}{6}$. Then $\mathbb{P}(\Omega) = 1$ and disjoint events add, so $\mathbb{P}$ is a probability measure. The event "even" has $\mathbb{P}(\{2, 4, 6\}) = \frac{3}{6} = \frac{1}{2}$.

## 2.6 Random Variables, Expectations and Variance

**Definition 2.21** (Random Variable)**.** *Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a probability space. A function $X : \Omega \to \mathbb{R}$ is called a (real-valued) random variable if it is $\mathcal{F}$-measurable, that is, for every Borel subset $\mathcal{B}$ of $\mathbb{R}$ the set*

$$\{X \in \mathcal{B}\} = \{\omega \in \Omega : X(\omega) \in \mathcal{B}\} \tag{19}$$

*belongs to $\mathcal{F}$.*

**Example 2.35** (The outcome of a die)**.** On the fair die space, let $X(\omega) = \omega$ report the face shown. It is measurable because the $\sigma$-algebra is the power set, so every $\{X \in \mathcal{B}\}$ lies in $\mathcal{F}$. A second example is the indicator $Y = \mathbf{1}_{\{2,4,6\}}$, the random variable equal to 1 on an even roll and 0 otherwise.

**Definition 2.22** (Distribution Measure)**.** *Let $X$ be a random variable on a probability space $(\Omega, \mathcal{F}, \mathbb{P})$. The distribution (or law) of $X$ is the pushforward measure $\mu_X := X_*\mathbb{P}$ on $(\mathbb{R}, \mathcal{B}[\mathbb{R}])$, explicitly, it assigns to each Borel set $\mathcal{B}$ the mass*

$$\mu_X(\mathcal{B}) = \mathbb{P}(X \in \mathcal{B}) = \mathbb{P}\big(X^{-1}(\mathcal{B})\big). \tag{20}$$

*Thus the distribution of a random variable is exactly the pushforward of the probability measure by that random variable.*

**Example 2.36.** Let $\mathbb{P}$ be the uniform measure on $[0, 1]$, where $\mathbb{P}[a, b] = b - a$ for $0 \le a \le b \le 1$. Define $X(\omega) = \omega$. Then $\mu_X[a, b] = \mathbb{P}[\{\omega : a \le \omega \le b\}] = b - a$, so the distribution of $X$ is again uniform on $[0, 1]$.

**Definition 2.23** (Expectation)**.** *Let $X$ be a random variable on $(\Omega, \mathcal{F}, \mathbb{P})$. If $X$ is $\mathbb{P}$-integrable, its expectation (or mean) is*

$$\mathbb{E}[X] = \int_\Omega X \, d\mathbb{P}. \tag{21}$$

*By the change-of-variables formula for pushforward measures, the expectation may equivalently be computed on the real line against the distribution of $X$, namely $\mathbb{E}[X] = \int_\mathbb{R} x \, d\mu_X(x)$, and more generally $\mathbb{E}[g(X)] = \int_\mathbb{R} g(x) \, d\mu_X(x)$ for every Borel $g$ with $g(X)$ integrable. The expectation is linear: $\mathbb{E}[aX + bY] = a\mathbb{E}[X] + b\mathbb{E}[Y]$.*

**Example 2.37** (Mean of a fair die)**.** For $X$ the face of a fair die, $\mathbb{E}[X] = \sum_{k=1}^{6} k \cdot \frac{1}{6} = \frac{1+2+3+4+5+6}{6} = 3.5$. The mean need not be an attainable value, it is the long-run average over many rolls.

The map $g \mapsto \mathbb{E}[g(X)]$ is positive (it sends non-negative $g$ to non-negative numbers) and normalised ($\mathbb{E}[1] = 1$). It is therefore a state on the algebra of bounded observables, and this is the classical prototype of the states that appear in the non-commutative theory.

**Definition 2.24** (Variance)**.** *Let $X$ be a random variable with finite mean $\mu = \mathbb{E}[X]$ and with $\mathbb{E}[X^2] < \infty$. The variance of $X$ is*

$$\mathrm{Var}(X) = \mathbb{E}\big[(X - \mu)^2\big] = \mathbb{E}[X^2] - \mu^2, \tag{22}$$

*a measure of the spread of $X$ about its mean.*

**Example 2.38** (Variance of a fair die)**.** For the fair die, $\mathbb{E}[X^2] = \sum_{k=1}^{6} k^2 \cdot \frac{1}{6} = \frac{91}{6}$, so $\mathrm{Var}(X) = \frac{91}{6} - (3.5)^2 = \frac{91}{6} - \frac{49}{4} = \frac{35}{12} \approx 2.92$. The larger the variance, the more the outcomes fluctuate around the mean of 3.5.

*Remark:.* The observation that $g \mapsto \mathbb{E}[g(X)]$ is a positive normalised functional is the entry point to the non-commutative theory. Passing from the commutative algebra of functions $g$ on $\mathbb{R}$ to a non-commutative $C^*$-algebra of observables, while retaining the notion of a positive normalised functional, is precisely the step from classical probability to quantum probability taken in Section 4.

# 3 Functional Analysis

Standard references for this section are Conway (8) and Reed and Simon (23), and for the operator-algebraic parts Murphy (19) and Bratteli and Robinson (6).

## 3.1 Vector Spaces and Norms

**Definition 3.1** (Vector Space)**.** *A vector space over a field $\mathbb{F}$ (where $\mathbb{F} = \mathbb{R}$ or $\mathbb{F} = \mathbb{C}$) is a non-empty set $V$ together with a binary operation $+ : V \times V \to V$ and a scalar multiplication $\cdot : \mathbb{F} \times V \to V$ satisfying, for every $\mathbf{u}, \mathbf{v}, \mathbf{w} \in V$ and $a, b \in \mathbb{F}$:*

1. $\mathbf{u} + (\mathbf{v} + \mathbf{w}) = (\mathbf{u} + \mathbf{v}) + \mathbf{w}$,
2. $\mathbf{u} + \mathbf{v} = \mathbf{v} + \mathbf{u}$,
3. *There exists* $\mathbf{0} \in V$ *such that* $\mathbf{v} + \mathbf{0} = \mathbf{v}$ *for all* $\mathbf{v} \in V$,
4. *For every* $\mathbf{v} \in V$, *there exists* $-\mathbf{v} \in V$ *such that* $\mathbf{v} + (-\mathbf{v}) = \mathbf{0}$,
5. $a(b\mathbf{v}) = (ab)\mathbf{v}$,
6. $1\mathbf{v} = \mathbf{v}$,
7. $a(\mathbf{u} + \mathbf{v}) = a\mathbf{u} + a\mathbf{v}$,
8. $(a + b)\mathbf{v} = a\mathbf{v} + b\mathbf{v}$.

**Example 3.1.** The set $\mathbb{R}^2$ with coordinatewise addition $(x_1, x_2) + (y_1, y_2) = (x_1 + y_1, x_2 + y_2)$ and scalar multiplication $a(x_1, x_2) = (ax_1, ax_2)$ is a vector space over $\mathbb{R}$, with zero vector $(0, 0)$. It is the familiar plane of arrows added tip to tail.

**Definition 3.2** (Norm)**.** *If $V$ is a vector space over $\mathbb{F} \in \{\mathbb{R}, \mathbb{C}\}$, then a norm on $V$ is a map $\|\cdot\| : V \to \mathbb{R}$ satisfying*

1. $\|x\| \geq 0$ *for every* $x \in V$,
2. $\|x\| = 0$ *if and only if* $x = 0$,
3. $\|\lambda x\| = |\lambda| \, \|x\|$ *for every* $\lambda \in \mathbb{F}$ *and* $x \in V$,
4. $\|x + y\| \leq \|x\| + \|y\|$ *for every* $x, y \in V$.

**Example 3.2.** On $\mathbb{R}^2$ the Euclidean norm $\|(x_1, x_2)\|_2 = \sqrt{x_1^2 + x_2^2}$ is a norm, for instance $\|(3, 4)\|_2 = 5$. It measures the ordinary length of a vector, and the triangle inequality says the length of a sum is at most the sum of the lengths.

A single vector space may carry many different norms. The following are among the most important.

**Example 3.3** ($\ell^p$ Norms on $\mathbb{F}^n$)**.** Let $x = (x_1, \ldots, x_n) \in \mathbb{F}^n$ and $1 \le p < \infty$. The $\ell^p$-norm is

$$\|x\|_p = \Big(\sum_{i=1}^{n} |x_i|^p\Big)^{1/p}. \tag{23}$$

The case $p = 1$ gives the taxicab norm $\|x\|_1 = \sum_i |x_i|$, and $p = 2$ gives the Euclidean norm. As $p \to \infty$ one obtains the supremum norm $\|x\|_\infty = \max_{1 \le i \le n} |x_i|$. For $x = (3, 4)$ we have $\|x\|_1 = 7$, $\|x\|_2 = 5$ and $\|x\|_\infty = 4$, showing that different norms assign different lengths to the same vector.

**Example 3.4** (Diagonal Norm)**.** Let $D = \operatorname{diag}(d_1, \ldots, d_n)$ with $d_i > 0$ for all $i$. The $D$-weighted norm on $\mathbb{F}^n$ is

$$\|x\|_D = \Big(\sum_{i=1}^{n} d_i |x_i|^2\Big)^{1/2} = \|D^{1/2} x\|_2. \tag{24}$$

This generalises the Euclidean norm, recovered when $D = I$. For $D = \operatorname{diag}(1, 4)$ and $x = (1, 1)$ we get $\|x\|_D = \sqrt{1 + 4} = \sqrt{5}$, so the second coordinate is weighted more heavily.

**Example 3.5** (Spectral Norm)**.** Let $A \in M_n(\mathbb{C})$ be self-adjoint, so $A = A^*$ and $A$ has real eigenvalues $\lambda_1, \ldots, \lambda_n$. The spectral norm of $A$ is $\|A\|_{\text{spec}} = \max_{1 \le i \le n} |\lambda_i|$. For a general matrix the spectral norm equals the largest singular value. For $A = \operatorname{diag}(2, -3)$ we get $\|A\|_{\text{spec}} = 3$.

**Example 3.6** (Trace Norm and Spectrum)**.** Let $A \in M_n(\mathbb{C})$. The spectrum of $A$ is $\sigma(A) = \{\lambda \in \mathbb{C} : \lambda I - A \text{ is not invertible}\}$. The trace norm is

$$\|A\|_1 = \operatorname{Tr}\big((A^* A)^{1/2}\big) = \sum_{i=1}^{n} \sigma_i(A), \tag{25}$$

where $\sigma_i(A)$ are the singular values. When $A \ge 0$ is positive semidefinite this reduces to $\|A\|_1 = \operatorname{Tr}(A)$. For $A = \operatorname{diag}(2, 3)$ we get $\|A\|_1 = 5$. The trace norm is the natural distance measure between density matrices later on.

## 3.2 From Normed Spaces to Metric Spaces

**Definition 3.3** (Normed Space)**.** *If $V$ is a vector space over $\mathbb{F}$ and $\|\cdot\|$ is a norm on $V$, then the pair $(V, \|\cdot\|)$ is called a normed space.*

**Example 3.7.** $(\mathbb{R}^n, \|\cdot\|_2)$ is a normed space, as is the space $C[0, 1]$ of continuous functions on $[0, 1]$ with $\|f\|_\infty = \max_{t \in [0,1]} |f(t)|$. In the latter, $\|f\|_\infty$ is the greatest height of the graph of $f$.

Every norm gives rise to a metric.

**Definition 3.4** (Metric Space)**.** *A metric space is an ordered pair $(M, d)$ where $M$ is a set and $d : M \times M \to [0, \infty)$ is a function satisfying, for all $x, y, z \in M$:*

1. *$d(x, x) = 0$,*
2. *If $x \ne y$, then $d(x, y) > 0$,*
3. *$d(x, y) = d(y, x)$,*

*4.* $d(x, z) \le d(x, y) + d(y, z)$.

**Example 3.8** (The discrete metric)**.** On any set $M$, define $d(x, y) = 0$ if $x = y$ and $d(x, y) = 1$ if $x \ne y$. This is a metric: every pair of distinct points sits at distance exactly 1. It shows that metrics need not come from norms and can be defined on sets with no algebraic structure.

**Proposition 3.1** (Induced Metric)**.** *Let $(V, \|\cdot\|)$ be a normed space. Then $d(x, y) := \|x - y\|$ defines a metric on $V$, called the metric induced by the norm.*

*Proof.* 1. $d(x, x) = \|x - x\| = \|0\| = 0$.

2. If $x \ne y$ then $x - y \ne 0$, so $d(x, y) = \|x - y\| > 0$ by definiteness of the norm.

3. $d(x, y) = \|x - y\| = \|-(y - x)\| = \|y - x\| = d(y, x)$.

4. $d(x, z) = \|(x - y) + (y - z)\| \le \|x - y\| + \|y - z\| = d(x, y) + d(y, z)$. □

**Example 3.9.** On $\mathbb{R}^2$ the Euclidean norm induces the straight-line distance $d(x, y) = \|x - y\|_2$, the distance from $(0, 0)$ to $(3, 4)$ is 5. Hence every normed space is automatically a metric space.

**Definition 3.5** (Complete Metric Space)**.** *A metric space $(M, d)$ is called complete if every Cauchy sequence in $M$ converges to a point in $M$.*

**Example 3.10.** $\mathbb{R}$ with $d(x, y) = |x - y|$ is complete: every Cauchy sequence of real numbers has a real limit. By contrast $\mathbb{Q}$ is not complete, since a sequence of rationals approaching $\sqrt{2}$ is Cauchy but has no rational limit.

**Definition 3.6** (Banach Space)**.** *A normed space $(V, \|\cdot\|)$ is called a Banach space if it is complete with respect to the induced metric $d(x, y) = \|x - y\|$.*

**Example 3.11.** $(\mathbb{R}^n, \|\cdot\|_2)$ is a Banach space, and so is $C[0, 1]$ with the supremum norm, because a uniformly Cauchy sequence of continuous functions converges uniformly to a continuous function. These complete normed spaces are the natural arenas of analysis.

### 3.3 Inner Product Spaces and Hilbert Spaces

**Definition 3.7** (Inner Product Space)**.** *An inner product space is a vector space $V$ over $\mathbb{F}$ together with a map $\langle \cdot, \cdot \rangle : V \times V \to \mathbb{F}$ such that for all $x, y, z \in V$ and $a, b \in \mathbb{F}$,*

*1.* $\langle x, y \rangle = \overline{\langle y, x \rangle}$ *(conjugate symmetry),*

*2.* $\langle ax + by, z \rangle = a\langle x, z \rangle + b\langle y, z \rangle$ *(linearity in the first argument),*

*3.* $\langle x, x \rangle \ge 0$*, with equality if and only if* $x = 0$ *(positive definiteness).*

*Conjugate symmetry forces $\langle x, x \rangle \in \mathbb{R}$, so axiom 3 is meaningful. Every inner product induces a norm via $\|x\| = \sqrt{\langle x, x \rangle}$ and hence, by Proposition 3.1, a metric.*

**Example 3.12.** On $\mathbb{C}^n$ the standard inner product is $\langle x, y \rangle = \sum_{i=1}^{n} x_i \overline{y_i}$. It is conjugate symmetric and linear in the first argument, and $\langle x, x \rangle = \sum_i |x_i|^2 = \|x\|_2^2$, so it induces the Euclidean norm. For $x = (1, i)$ we get $\langle x, x \rangle = 1 + 1 = 2$.

**Theorem 3.1** (Cauchy–Schwarz Inequality). *Let $(V, \langle \cdot, \cdot \rangle)$ be an inner product space. Then for all $x, y \in V$,*

$$|\langle x, y \rangle| \leq \|x\| \, \|y\|, \tag{26}$$

*with equality if and only if $x$ and $y$ are linearly dependent.*

**Example 3.13.** On $\mathbb{R}^2$ with the dot product, take $x = (1, 0)$ and $y = (1, 1)$. Then $|\langle x, y \rangle| = 1$ while $\|x\| \, \|y\| = 1 \cdot \sqrt{2} = \sqrt{2}$, so indeed $1 \leq \sqrt{2}$. Equality would require $x$ and $y$ to be parallel, which they are not.

**Definition 3.8** (Hilbert Space). *A Hilbert space is an inner product space $(V, \langle \cdot, \cdot \rangle)$ that is complete with respect to the norm $\|x\| = \sqrt{\langle x, x \rangle}$. Every Hilbert space is in particular a Banach space.*

**Example 3.14.** $\mathbb{C}^n$ with the standard inner product is a Hilbert space. An infinite-dimensional example is $\ell^2 = \{(x_n) : \sum_n |x_n|^2 < \infty\}$ with $\langle x, y \rangle = \sum_n x_n \overline{y_n}$, it is complete, and it is the model space on which the quantum-mechanical formalism of the later sections is built.

### 3.4 Linear Operators on Normed Spaces

**Definition 3.9** (Linear Operator). *A map $T : V \to W$ between two vector spaces $V$ and $W$ over $\mathbb{F}$ is called a linear operator if for all $v_1, v_2 \in V$ and $\lambda \in \mathbb{F}$,*

1. *$T(\lambda v_1) = \lambda T(v_1)$,*
2. *$T(v_1 + v_2) = T(v_1) + T(v_2)$.*

*We denote the set of all linear operators from $V$ to $W$ by $L(V, W)$.*

**Example 3.15.** The map $T : \mathbb{R}^2 \to \mathbb{R}^2$ given by $T(x_1, x_2) = (2x_1,\, x_1 + x_2)$ is linear, it is represented by the matrix $\left(\begin{smallmatrix} 2 & 0 \\ 1 & 1 \end{smallmatrix}\right)$. Differentiation $f \mapsto f'$ on the space of polynomials is another linear operator, since the derivative of a sum is the sum of derivatives.

**Definition 3.10** (Bounded Linear Operator). *Let $X$ and $Y$ be normed spaces and $T \in L(X, Y)$. Then $T$ is called bounded if there exists a constant $\lambda \geq 0$ such that*

$$\|Tx\|_Y \leq \lambda \|x\|_X \quad \text{for all } x \in X. \tag{27}$$

*The set of all bounded linear operators from $X$ to $Y$ is denoted $B(X, Y)$.*

In finite dimensions every linear operator is bounded, since the unit sphere is compact and the norm is continuous. In infinite dimensions boundedness is a genuine restriction.

**Example 3.16.** The operator $T(x_1, x_2) = (2x_1, x_1 + x_2)$ is bounded because $\|Tx\|_2 \leq 3\|x\|_2$ for all $x$. An unbounded example is differentiation on $C^1[0, 1] \subseteq C[0, 1]$ with the supremum norm: $f_n(t) = \sin(nt)$ has $\|f_n\|_\infty = 1$ but $\|f_n'\|_\infty = n \to \infty$, so no single constant $\lambda$ works.

**Definition 3.11** (Operator Norm). *Let $X$ and $Y$ be normed spaces. For $T \in B(X, Y)$, the operator norm of $T$ is*

$$\|T\|_{B(X,Y)} = \sup_{\substack{x \in X \\ x \neq 0}} \frac{\|Tx\|_Y}{\|x\|_X} = \sup_{\|x\|_X \leq 1} \|Tx\|_Y. \tag{28}$$

**Example 3.17.** For $T = \left(\begin{smallmatrix} 2 & 0 \\ 0 & 3 \end{smallmatrix}\right)$ on $(\mathbb{R}^2, \|\cdot\|_2)$, the image of the unit circle is an ellipse with semi-axes 2 and 3, so $\|T\| = 3$, the largest stretch factor. In general the operator norm of a matrix is its largest singular value.

**Proposition 3.2.** *The operator norm $\|\cdot\|_{B(X,Y)}$ is a norm on $B(X,Y)$. Moreover, if $Y$ is a Banach space then $B(X,Y)$ is itself a Banach space.*

*Proof.* Let $T, S \in B(X,Y)$ and $\alpha \in \mathbb{F}$.

1. Non-negativity. $\|T\| \geq 0$ since $\|Tx\|_Y \geq 0$ for all $x$.
2. Definiteness. If $\|T\| = 0$ then $\|Tx\| = 0$ for all $\|x\| \leq 1$, so $Tx = 0$ for all $x$, that is $T = 0$.
3. Homogeneity. $\|\alpha T\| = \sup_{\|x\|\leq 1} \|\alpha Tx\| = |\alpha| \sup_{\|x\|\leq 1} \|Tx\| = |\alpha|\|T\|$.
4. Triangle inequality. $\|(T+S)x\| \leq \|Tx\| + \|Sx\| \leq (\|T\| + \|S\|)\|x\|$, so $\|T+S\| \leq \|T\| + \|S\|$.

Completeness of $B(X,Y)$ when $Y$ is complete is assumed. $\square$

**Example 3.18.** On $(\mathbb{R}^2, \|\cdot\|_2)$ take $T = \left(\begin{smallmatrix} 2 & 0 \\ 0 & 0 \end{smallmatrix}\right)$ and $S = \left(\begin{smallmatrix} 0 & 0 \\ 0 & 3 \end{smallmatrix}\right)$, with $\|T\| = 2$ and $\|S\| = 3$. Then $T + S = \left(\begin{smallmatrix} 2 & 0 \\ 0 & 3 \end{smallmatrix}\right)$ has $\|T+S\| = 3 \leq 2 + 3$, illustrating the triangle inequality for the operator norm.

### 3.5 Dual Spaces

**Definition 3.12** (Dual Space)**.** *Let $V$ be a vector space over $\mathbb{F}$. The dual space of $V$, denoted $V^* = L(V, \mathbb{F})$, is the set of all linear functionals $\varphi : V \to \mathbb{F}$, a vector space under pointwise operations $(\varphi + \psi)(x) = \varphi(x) + \psi(x)$ and $(a\varphi)(x) = a\,\varphi(x)$.*

**Example 3.19.** On $\mathbb{R}^n$ every linear functional has the form $\varphi(x) = \sum_i c_i x_i$ for a fixed row vector $(c_1, \ldots, c_n)$. Thus $(\mathbb{R}^n)^*$ is again $n$-dimensional, for example $\varphi(x_1, x_2) = 3x_1 - x_2$ is a functional on $\mathbb{R}^2$.

**Definition 3.13** (Normed Dual Space)**.** *Let $(V, \|\cdot\|)$ be a normed space. The normed dual of $V$ is $V' = B(V, \mathbb{F}) = \{\varphi \in V^* : \varphi \text{ is bounded}\}$, equipped with the operator norm $\|\varphi\|_{V'} = \sup_{\|x\|\leq 1} |\varphi(x)|$. Since $\mathbb{F}$ is complete, $V'$ is always a Banach space. For every $\varphi \in V'$ and $x \in V$, $|\varphi(x)| \leq \|\varphi\|_{V'} \|x\|_V$.*

The normed dual generalises the transpose to infinite dimensions and plays a central role in the Kantorovich duality for optimal transport, where the transport cost is paired against continuous test functions.

**Example 3.20.** On $C[0,1]$ the evaluation $\varphi(f) = f(\frac{1}{2})$ is a bounded functional with $\|\varphi\|_{V'} = 1$, since $|f(\frac{1}{2})| \leq \|f\|_\infty$ and equality holds for the constant function 1. Integration $f \mapsto \int_0^1 f(t)\,dt$ is another bounded functional, also of norm 1.

**Theorem 3.2** (Riesz Representation Theorem for Hilbert Spaces)**.** *Let $\mathcal{H}$ be a Hilbert space. For every bounded linear functional $\varphi \in \mathcal{H}'$ there exists a unique $y \in \mathcal{H}$ such that $\varphi(x) = \langle x, y \rangle$ for all $x \in \mathcal{H}$, and moreover $\|\varphi\|_{\mathcal{H}'} = \|y\|$.*

**Example 3.21.** On $\mathbb{R}^2$ the functional $\varphi(x) = 3x_1 - x_2$ is represented by the vector $y = (3, -1)$, because $\varphi(x) = \langle x, (3, -1) \rangle$. The theorem says every functional on a Hilbert space is "take the inner product with a fixed vector", identifying $\mathcal{H}$ with its own dual.

### 3.6 Tensor Products

The tensor product is most cleanly described by a universal property that requires no choice of basis and applies in both finite and infinite dimensions.

**Definition 3.14** (Bilinear Map)**.** *Let $U, V, W$ be vector spaces over $\mathbb{F}$. A map $\beta : U \times V \to W$ is called bilinear if it is linear in each argument:*

$$\beta(au_1 + bu_2, v) = a\beta(u_1, v) + b\beta(u_2, v), \quad \beta(u, av_1 + bv_2) = a\beta(u, v_1) + b\beta(u, v_2), \tag{29}$$

*for all $u, u_1, u_2 \in U$, $v, v_1, v_2 \in V$, and $a, b \in \mathbb{F}$.*

**Example 3.22.** The dot product $\beta(u, v) = u \cdot v$ on $\mathbb{R}^2 \times \mathbb{R}^2$ is bilinear, as is ordinary multiplication $\beta(a, b) = ab$ on $\mathbb{R} \times \mathbb{R}$. In each case fixing one argument leaves a linear function of the other.

**Definition 3.15** (Isomorphic Functions)**.** *Let $f : X \to Y$ and $g : A \to B$ be functions. An isomorphism between $f$ and $g$ is a pair of bijections $\phi : X \to A$ and $\psi : Y \to B$ such that $\psi \circ f = g \circ \phi$, that is, $\psi(f(x)) = g(\phi(x))$ for every $x \in X$. This is exactly the statement that the following square commutes:*

$$\begin{tikzcd} X \arrow[r, "f"] \arrow[d, "\phi"'] & Y \arrow[d, "\psi"] \\ A \arrow[r, "g"] & B \end{tikzcd}$$

**Definition 3.16** (Factorization of a Function)**.** *Let $f : X \to Y$, $\rho : Y \to Z$, and $g : X \to Z$ be functions. We say that $g$ factors through $f$ via $\rho$ if $g = \rho \circ f$, that is, $g(x) = \rho(f(x))$ for every $x \in X$. Equivalently, the following triangle commutes:*

$$\begin{tikzcd} X \arrow[r, "f"] \arrow[dr, "g"'] & Y \arrow[d, "\rho"] \\ & Z \end{tikzcd}$$

**Example 3.23.** Let $f : \mathbb{R} \to \mathbb{R}$ be $f(x) = x + 1$ and $g : \mathbb{R} \to \mathbb{R}$ be $g(x) = 2x$. With $\phi(x) = 2x$ and $\psi(y) = 2y - 2$ one checks $\psi(f(x)) = 2(x + 1) - 2 = 2x = g(\phi(x))$, so $f$ and $g$ are isomorphic: relabelling the inputs and outputs turns "add one" into "double".

**Theorem 3.3.** *Let $f : X \to Y$ and $g : X \to Z$ be functions. The following are equivalent:*

1. *The triangle*

$$\begin{tikzcd} X \arrow[r, "f"] \arrow[dr, "g"'] & Y \arrow[d, "\rho"] \\ & Z \end{tikzcd}$$

   *commutes.*

2. *There exists a function $\rho : Y \to Z$ such that $g = \rho \circ f$.*

**Theorem 3.4** (Universal Property of the Tensor Product)**.** *Let $V$ and $W$ be vector spaces over $\mathbb{F}$. There exists a vector space $V \otimes W$, together with a bilinear map $\otimes : V \times W \to V \otimes W$, $(v, w) \mapsto v \otimes w$, satisfying the following universal property: for every vector space $Z$ and every bilinear map $\beta : V \times W \to Z$, there exists a unique linear map $\widetilde{\beta} : V \otimes W \to Z$ such that $\widetilde{\beta}(v \otimes w) = \beta(v, w)$ for all $v \in V$, $w \in W$. The pair $(V \otimes W, \otimes)$ is unique up to isomorphism.*

**Example 3.24.** The bilinear map $\beta(u, v) = u \cdot v$ on $\mathbb{R}^2 \times \mathbb{R}^2$ factors through the tensor product as a unique linear map $\widetilde{\beta} : \mathbb{R}^2 \otimes \mathbb{R}^2 \to \mathbb{R}$ with $\widetilde{\beta}(u \otimes v) = u \cdot v$. Turning a bilinear operation into a single linear map on the tensor product is exactly what the universal property provides.

**Definition 3.17** (Elementary Tensor)**.** *An element of the form $v \otimes w \in V \otimes W$, for $v \in V$ and $w \in W$, is called an elementary tensor. The space $V \otimes W$ is spanned by elementary tensors, but not every element is elementary, a general element is a finite sum $\sum_i a_i(v_i \otimes w_i)$.*

**Example 3.25.** In $\mathbb{R}^2 \otimes \mathbb{R}^2$ the element $e_1 \otimes e_1$ is elementary, whereas $e_1 \otimes e_1 + e_2 \otimes e_2$ is not: it cannot be written as a single $v \otimes w$. In the quantum setting such non-elementary tensors are exactly the entangled states.

**Theorem 3.5** (Properties of Tensor Products)**.** *Let $V, W$ be vector spaces over $\mathbb{F}$. For all $v, v_1, v_2 \in V$, $w, w_1, w_2 \in W$, and $z \in \mathbb{F}$:*

1. $z(v \otimes w) = (zv) \otimes w = v \otimes (zw)$,
2. $(v_1 + v_2) \otimes w = v_1 \otimes w + v_2 \otimes w$,
3. $v \otimes (w_1 + w_2) = v \otimes w_1 + v \otimes w_2$.

**Example 3.26.** In $\mathbb{R}^2 \otimes \mathbb{R}^2$, property 2 gives $(e_1 + e_2) \otimes e_1 = e_1 \otimes e_1 + e_2 \otimes e_1$, and property 1 gives $3(e_1 \otimes e_2) = (3e_1) \otimes e_2 = e_1 \otimes (3e_2)$, showing that scalars may be moved across the tensor sign.

The tensor product extends to linear operators. If $A : V \to V'$ and $B : W \to W'$ are linear, the universal property yields a unique linear map $A \otimes B : V \otimes W \to V' \otimes W'$ with $(A \otimes B)(v \otimes w) = Av \otimes Bw$. If $V$ and $W$ are inner product spaces, then $V \otimes W$ inherits an inner product with $\langle v_1 \otimes w_1, v_2 \otimes w_2 \rangle = \langle v_1, v_2 \rangle \langle w_1, w_2 \rangle$.

**Theorem 3.6** (Basis of a Tensor Product Space)**.** *Let $V$ and $W$ be finite-dimensional with bases $\mathcal{B}_V = \{v_1, \ldots, v_m\}$ and $\mathcal{B}_W = \{w_1, \ldots, w_n\}$. Then $\mathcal{B}_{V \otimes W} = \{v_i \otimes w_j : 1 \le i \le m,\ 1 \le j \le n\}$ is a basis for $V \otimes W$, so $\dim(V \otimes W) = \dim(V)\dim(W)$.*

**Example 3.27.** Let $V = \mathbb{R}^2$ and $W = \mathbb{R}^3$ with standard bases $e_1, e_2$ and $f_1, f_2, f_3$. Then $V \otimes W = \operatorname{span}\{e_i \otimes f_j\}$ has the six basis elements $e_1 \otimes f_1, \ldots, e_2 \otimes f_3$, so $\dim(V \otimes W) = 6 = 2 \cdot 3$.

Once bases are chosen, the matrix of $A \otimes B$ is the Kronecker product.

**Definition 3.18** (Kronecker Product)**.** *Let $A \in M_{m \times n}(\mathbb{F})$ and $B \in M_{p \times q}(\mathbb{F})$. The Kronecker product of $A$ and $B$ is the block matrix*

$$A \otimes B = \begin{bmatrix} A_{11}B & A_{12}B & \cdots & A_{1n}B \\ A_{21}B & A_{22}B & \cdots & A_{2n}B \\ \vdots & \vdots & \ddots & \vdots \\ A_{m1}B & A_{m2}B & \cdots & A_{mn}B \end{bmatrix}, \tag{30}$$

*an $(mp) \times (nq)$ matrix.*

**Example 3.28.** Let $A = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$ and $B = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$. Then

$$A \otimes B = \begin{pmatrix} 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & -1 \\ 0 & 0 & -1 & 0 \end{pmatrix}. \tag{31}$$

These are the Pauli matrices $Z$ and $X$, and $Z \otimes X$ is the operator acting on a two-qubit system.

**Example 3.29.** For $A = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$ and $B = \begin{pmatrix} e & f \\ g & h \end{pmatrix}$,

$$A \otimes B = \begin{bmatrix} ae & af & be & bf \\ ag & ah & bg & bh \\ ce & cf & de & df \\ cg & ch & dg & dh \end{bmatrix}. \tag{32}$$

**Definition 3.19** (Tensor Product of Functions)**.** *Suppose $f : X \to \mathbb{R}$ and $g : Y \to \mathbb{R}$. Their tensor product is the function $(f \otimes g)(x, y) = f(x)g(y)$ on $X \times Y$.*

**Example 3.30.** Let $f(x) = x^2$ and $g(y) = \sin(y)$. Then $(f \otimes g)(x, y) = x^2 \sin(y)$, a function of two variables built by multiplying a function of $x$ by a function of $y$.

When the domains are finite, functions are vectors and the tensor product is the Kronecker product.

**Example 3.31** (Tensor Product via the Kronecker Product)**.** Let $X = \{x_1, x_2\}$, $Y = \{y_1, y_2, y_3\}$, and define $f, g$ by $f(x_1) = 1$, $f(x_2) = 2$, $g(y_1) = 3$, $g(y_2) = 4$, $g(y_3) = 5$. As column vectors,

$$f = \begin{bmatrix} 1 \\ 2 \end{bmatrix}, \qquad g = \begin{bmatrix} 3 \\ 4 \\ 5 \end{bmatrix}, \qquad f \otimes g = \begin{bmatrix} 1 \cdot g \\ 2 \cdot g \end{bmatrix} = \begin{bmatrix} 3 \\ 4 \\ 5 \\ 6 \\ 8 \\ 10 \end{bmatrix}, \tag{33}$$

where each entry is $f(x_i)g(y_j)$.

# 4 Optimal Transport Theory

Suppose we have two probability measures $\mu$ and $\nu$ defined on $X$ and $Y$ respectively. With optimal transport we wish to move one measure, say $\mu$, onto the other, $\nu$, at least cost. The cost is a function $c : X \times Y \to [0, +\infty]$, where $c(x, y)$ is the cost of transporting one unit of mass from $x$ to $y$. The aim is to minimise the total cost while transporting $\mu$ to $\nu$. Standard references are Villani (28, 29), Santambrogio (26) and Ambrosio, Gigli and Savaré (2).

## 4.1 The Monge Problem and the Kantorovich Relaxation

**Definition 4.1** (Transport Map)**.** *Let $(X, \mu)$, $(Y, \nu)$ be probability spaces. A measurable map $T : X \to Y$ is called a transport map from $\mu$ to $\nu$ if $\nu = T_*\mu$, that is, $\nu(B) = \mu(T^{-1}(B))$ for all measurable $B \subseteq Y$.*

**Example 4.1** (A translation)**.** Let $\mu$ be uniform on $[0,1]$ and $\nu$ uniform on $[1,2]$, and let $T(x) = x+1$. For an interval $[c,d] \subseteq [1,2]$, $T^{-1}([c,d]) = [c-1, d-1]$, so $T_*\mu([c,d]) = d - c = \nu([c,d])$. Thus $T$ is a transport map: it slides every point of $[0,1]$ one unit to the right.

**Definition 4.2** (Transport Plan)**.** *Let $(A,\mu)$ and $(B,\nu)$ be probability spaces. A transport plan (or coupling) from $\mu$ to $\nu$ is a probability measure $\pi \in \mathcal{P}(A \times B)$ whose marginals are $\mu$ and $\nu$, that is,*

$$\int_{A\times B} f(a)\, d\pi(a,b) = \int_A f(a)\, d\mu(a) \qquad \textit{and} \qquad \int_{A\times B} g(b)\, d\pi(a,b) = \int_B g(b)\, d\nu(b) \tag{34}$$

*for all bounded measurable $f : A \to \mathbb{R}$ and $g : B \to \mathbb{R}$. Writing $\mu(f) = \int f\, d\mu$ and $(f \otimes g)(a,b) = f(a)g(b)$, the marginal conditions read $\pi(f \otimes 1) = \mu(f)$ and $\pi(1 \otimes g) = \nu(g)$.*

The set of all transport plans between $\mu$ and $\nu$ is denoted

$$\Pi(\mu,\nu) = \{\pi \in \mathcal{P}(A \times B) : \pi_A = \mu,\ \pi_B = \nu\}, \tag{35}$$

where $\pi_A, \pi_B$ are the marginals of $\pi$. This set is never empty, because the product measure $\mu \otimes \nu$ (the independent coupling) always belongs to it. The functional form $\pi(f \otimes 1) = \mu(f)$, $\pi(1 \otimes g) = \nu(g)$ is the one that generalises verbatim to the non-commutative setting.

**Example 4.2** (A discrete coupling, and the classical to quantum crossing point)**.** Take $A = \{1,2\}$ and $B = \{1,2\}$, with probability vectors $\mu = (0.6, 0.4)$ and $\nu = (0.5, 0.5)$. A transport plan is a joint distribution matrix

$$\pi = \begin{pmatrix} p_{11} & p_{12} \\ p_{21} & p_{22} \end{pmatrix} = \begin{pmatrix} 0.3 & 0.3 \\ 0.2 & 0.2 \end{pmatrix}, \qquad \sum_j p_{ij} = \mu_i, \quad \sum_i p_{ij} = \nu_j. \tag{36}$$

The row sums are $(0.6, 0.4) = \mu$ and the column sums are $(0.5, 0.5) = \nu$, so $\pi \in \Pi(\mu,\nu)$. This is also where the classical and quantum pictures meet. A probability vector on $A$ is a state on the commutative algebra $\mathbb{C}^2$ of functions on $A$, realised as the diagonal subalgebra of $M_2(\mathbb{C})$, likewise for $B$. The plan $\pi$ is a state on $\mathbb{C}^2 \otimes \mathbb{C}^2 \cong \mathbb{C}^4$, that is, on the diagonal subalgebra of $M_2(\mathbb{C}) \otimes M_2(\mathbb{C})$. A genuine quantum coupling replaces these diagonal algebras by the full matrix algebras. Classical couplings are the special case in which every operator is diagonal, the off-diagonal entries are what is new in the non-commutative theory.

**Definition 4.3** (Monge's Optimal Transport Problem)**.** *Let $(X,\mu)$, $(Y,\nu)$ be probability spaces and $c : X \times Y \to [0,+\infty]$ a cost. Monge's problem is*

$$\textit{minimise } \mathbb{M}(T) = \int_X c(x, T(x))\, d\mu(x) \tag{37}$$

*over measurable maps $T : X \to Y$ with $\nu = T_*\mu$.*

**Example 4.3.** With $\mu$ uniform on $[0,1]$, $\nu$ uniform on $[1,2]$, cost $c(x,y) = |x-y|$, and $T(x) = x+1$, the Monge cost is $\mathbb{M}(T) = \int_0^1 |x - (x+1)|\, dx = \int_0^1 1\, dx = 1$. Every unit of mass travels distance 1, so the total cost is 1.

**Definition 4.4** (Kantorovich's Optimal Transport Problem)**.** *Let $(X,\mu)$, $(Y,\nu)$ be probability spaces and $c : X \times Y \to [0,+\infty]$ a cost. Kantorovich's problem is*

$$\textit{minimise } \mathbb{K}(\pi) = \int_{X\times Y} c(x,y)\, d\pi(x,y) \tag{38}$$

*over $\pi \in \Pi(\mu,\nu)$.*

**Example 4.4** (Kantorovich as a small linear program)**.** Let $\mu = (0.6, 0.4)$ on $\{1, 2\}$ and $\nu = (0.5, 0.5)$ on $\{1, 2\}$ with cost matrix $c = \left(\begin{smallmatrix} 0 & 1 \\ 1 & 0 \end{smallmatrix}\right)$. Minimising $\sum_{ij} c_{ij} p_{ij} = p_{12} + p_{21}$ over couplings with the prescribed margins gives the optimum $\pi = \left(\begin{smallmatrix} 0.5 & 0.1 \\ 0 & 0.4 \end{smallmatrix}\right)$ with cost 0.1: as much mass as possible stays on the diagonal (cost 0), and only the unavoidable 0.1 is moved. This is a finite linear program, which is exactly why Kantorovich's relaxation is tractable.

**Definition 4.5** (Diagonal Measure)**.** *Let $(X, \mu)$ be a measure space. The diagonal map $\Delta : X \to X \times X$ is $\Delta(x) = (x, x)$, and the diagonal measure induced by $\mu$ is the pushforward $\Delta_{\#}\mu \in \mathcal{P}(X \times X)$. Explicitly, for measurable $E \subseteq X \times X$, $(\Delta_{\#}\mu)(E) = \mu(\{x : (x, x) \in E\})$.*

The diagonal measure is the no-transport coupling: all mass at $x$ stays at $x$.

**Example 4.5.** For $\mu = \frac{1}{2}(\delta_0 + \delta_1)$ on $\{0, 1\}$, the diagonal measure is $\Delta_{\#}\mu = \frac{1}{2}\delta_{(0,0)} + \frac{1}{2}\delta_{(1,1)}$, which couples each point only with itself. Its cost under any $c$ with $c(x, x) = 0$ is zero, so it is optimal whenever $\mu = \nu$.

## 4.2 Relationship Between the Monge and Kantorovich Problems

Every transport map induces a transport plan, so Kantorovich's problem is a relaxation of Monge's. If $T$ is a transport map, then $\pi_T := (\mathrm{id} \times T)_* \mu \in \Pi(\mu, \nu)$ is a coupling concentrated on the graph of $T$, and

$$\mathbb{K}(\pi_T) = \int_{X \times Y} c(x, y)\, d\pi_T(x, y) = \int_X c(x, T(x))\, d\mu(x) = \mathbb{M}(T). \tag{39}$$

Consequently $\inf_{\pi \in \Pi(\mu,\nu)} \mathbb{K}(\pi) \leq \inf_{T : T_*\mu = \nu} \mathbb{M}(T)$. Two features distinguish the problems. First, the Monge problem may be infeasible while the Kantorovich problem is not: if $\mu = \delta_a$ and $\nu = \frac{1}{2}(\delta_b + \delta_c)$ with $b \neq c$, no map can split the mass at $a$, yet $\Pi(\mu, \nu)$ is non-empty. Second, the Kantorovich problem, being linear in $\pi$ over the weak-$*$ compact convex set $\Pi(\mu, \nu)$, always attains its infimum when $c$ is lower semicontinuous.

**Example 4.6** (Splitting mass)**.** Let $\mu = \delta_0$ and $\nu = \frac{1}{2}(\delta_{-1} + \delta_1)$ with cost $c(x, y) = |x - y|$. No transport map exists, since a function cannot send the single point 0 to two places. The unique coupling is $\pi = \frac{1}{2}\delta_{(0,-1)} + \frac{1}{2}\delta_{(0,1)}$, with Kantorovich cost $\frac{1}{2} \cdot 1 + \frac{1}{2} \cdot 1 = 1$. Thus Kantorovich succeeds precisely where Monge has no feasible point.

**Theorem 4.1** (Kantorovich Duality)**.** *Let $X, Y$ be Polish spaces, $\mu \in \mathcal{P}(X)$, $\nu \in \mathcal{P}(Y)$, and let $c : X \times Y \to [0, +\infty]$ be lower semicontinuous. Then*

$$\min_{\pi \in \Pi(\mu,\nu)} \int_{X \times Y} c\, d\pi = \sup_{(\varphi,\psi) \in \Phi_c} \left( \int_X \varphi\, d\mu + \int_Y \psi\, d\nu \right), \tag{40}$$

*where $\Phi_c = \{(\varphi, \psi) \in L^1(\mu) \times L^1(\nu) : \varphi(x) + \psi(y) \leq c(x, y)$ for all $x, y\}$.*

The pair $(\varphi, \psi)$ are the Kantorovich potentials, and $\varphi(x) + \psi(y) \leq c(x, y)$ is the dual feasibility condition. This is where the dual-space theory of Section 3 enters. When the cost is a metric, $c(x, y) = d(x, y)$, the potentials can be taken equal and 1-Lipschitz, giving the Kantorovich–Rubinstein formula

$$W_1(\mu, \nu) = \sup_{\mathrm{Lip}(f) \leq 1} \left( \int_X f\, d\mu - \int_X f\, d\nu \right). \tag{41}$$

**Example 4.7** (Duality for two points)**.** Take $\mu = \delta_0$, $\nu = \delta_1$ on $\mathbb{R}$ with $c(x, y) = |x - y|$. The only coupling is $\delta_{(0,1)}$, of cost 1, so the primal minimum is 1. On the dual side, choose the 1-Lipschitz function $f(x) = x$, then $\int f\, d\mu - \int f\, d\nu = 0 - 1 = -1$, and $f(x) = -x$ gives $+1$. The dual supremum is 1, matching the primal, in agreement with the theorem.

**Theorem 4.2** (Brenier's Theorem)**.** *Let $\mu, \nu \in \mathcal{P}_2(\mathbb{R}^n)$ with $\mu$ absolutely continuous with respect to Lebesgue measure, and take the quadratic cost $c(x, y) = \frac{1}{2}|x - y|^2$. Then the Kantorovich problem has a unique optimal plan, it is induced by a transport map $T$ (so Monge and Kantorovich coincide), and $T = \nabla\phi$ for some convex function $\phi : \mathbb{R}^n \to \mathbb{R}$.*

**Example 4.8.** Let $\mu$ be uniform on $[0, 1]$ and $\nu$ uniform on $[1, 2]$. The convex potential $\phi(x) = \frac{1}{2}x^2 + x$ has gradient $\phi'(x) = x + 1 = T(x)$, which is precisely the optimal (translation) map. Brenier's theorem thus recovers the intuitive "shift by one" map as the gradient of a convex function.

### 4.3 The Wasserstein $W_p$ Metric

Fix $p \in [1, \infty)$ and a basepoint $x_0 \in X$. The space of probability measures with finite $p$-th moment is

$$\mathcal{P}_p(X) = \left\{ \mu \in \mathcal{P}(X) : \int_X d(x_0, x)^p\, d\mu(x) < \infty \right\}, \tag{42}$$

which does not depend on the choice of $x_0$.

**Definition 4.6** (Wasserstein-$p$ Distance)**.** *Let $(X, d)$ be a Polish space, let $p \in [1, \infty)$, and let $\mu, \nu \in \mathcal{P}_p(X)$. The Wasserstein-$p$ distance between $\mu$ and $\nu$ is*

$$W_p(\mu, \nu) := \left( \inf_{\pi \in \Pi(\mu,\nu)} \int_{X \times X} d(x, y)^p\, d\pi(x, y) \right)^{1/p}. \tag{43}$$

*The infimum is attained, since the cost is lower semicontinuous and $\Pi(\mu, \nu)$ is weak-$*$ compact.*

The requirement $p \geq 1$ makes $W_p$ satisfy the triangle inequality. On $\mathcal{P}_p(X)$, $W_p$ is a genuine metric: symmetry is clear, and $W_p(\mu, \nu) = 0$ forces $\mu = \nu$, because the only zero-cost coupling is the diagonal measure. Moreover $W_p$ metrises weak convergence together with convergence of $p$-th moments (29).

**Example 4.9** (Two Dirac masses)**.** For $\mu = \delta_a$ and $\nu = \delta_b$ on $\mathbb{R}$, the only coupling is $\delta_{(a,b)}$, so $W_p(\delta_a, \delta_b) = (|a - b|^p)^{1/p} = |a - b|$ for every $p$. On point masses the Wasserstein distance is just the ground distance between the two points.

**Example 4.10** (Wasserstein distance between discrete measures on $\mathbb{R}$)**.** Consider $\mu = \frac{1}{2}(\delta_0 + \delta_2)$ and $\nu = \frac{1}{2}(\delta_1 + \delta_3)$ on $(\mathbb{R}, |\cdot|)$. Since both measures put mass $\frac{1}{2}$ on two source points $\{0, 2\}$ and two target points $\{1, 3\}$, there are exactly two couplings,

$$\pi_1 = \tfrac{1}{2}\delta_{(0,1)} + \tfrac{1}{2}\delta_{(2,3)}, \qquad \pi_2 = \tfrac{1}{2}\delta_{(0,3)} + \tfrac{1}{2}\delta_{(2,1)}, \tag{44}$$

with costs $\int |x - y|^p\, d\pi_1 = \frac{1}{2}\, 1^p + \frac{1}{2}\, 1^p = 1$ and $\int |x - y|^p\, d\pi_2 = \frac{1}{2}\, 3^p + \frac{1}{2}\, 1^p \geq 1$. The non-crossing coupling $\pi_1$ is optimal for every $p \geq 1$, so $W_p(\mu, \nu) = 1$. This is the smallest instance of the one-dimensional formula $W_p(\mu, \nu)^p = \int_0^1 |F_\mu^{-1}(t) - F_\nu^{-1}(t)|^p\, dt$ in terms of quantile functions. For $p = 1$ it reads $W_1(\mu, \nu) = \int_{\mathbb{R}} |F_\mu(t) - F_\nu(t)|\, dt$, the area between the two cumulative distribution functions. Since actuarial risk measures such as Value-at-Risk $\mathrm{VaR}_\alpha = F^{-1}(\alpha)$ and Tail-Value-at-Risk $\mathrm{TVaR}_\alpha = \frac{1}{1-\alpha} \int_\alpha^1 F^{-1}$ are functionals of the quantile function, the Wasserstein distance controls their sensitivity to distributional perturbations, a link exploited in Wasserstein distributionally robust optimisation (18).

## 4.4 Operator Algebras

**Definition 4.7** (Algebra)**.** *An algebra $\mathcal{A}$ is a vector space over $\mathbb{C}$ equipped with a product $\cdot : (a, b) \mapsto ab$ satisfying, for all $a, b, c \in \mathcal{A}$ and $\alpha, \beta \in \mathbb{C}$,*

1. $a \cdot (b \cdot c) = (a \cdot b) \cdot c$ *(associativity),*
2. $a \cdot (b + c) = a \cdot b + a \cdot c$ *and* $(b + c) \cdot a = b \cdot a + c \cdot a$ *(left and right distributivity),*
3. $(\alpha a) \cdot (\beta b) = (\alpha\beta)(a \cdot b)$ *(bilinearity of the product).*

*Since the algebra need not be commutative, both distributive laws must be assumed. If there exists $1 \in \mathcal{A}$ with $1 \cdot a = a \cdot 1 = a$ for all $a$, then $\mathcal{A}$ is unital.*

**Example 4.11.** The set $M_n(\mathbb{C})$ of $n \times n$ complex matrices, with matrix addition and multiplication, is a unital algebra, it is non-commutative for $n \geq 2$, since $\left(\begin{smallmatrix} 0 & 1 \\ 0 & 0 \end{smallmatrix}\right)$ and $\left(\begin{smallmatrix} 0 & 0 \\ 1 & 0 \end{smallmatrix}\right)$ do not commute. The set $C(X)$ of continuous complex functions on a compact space $X$, with pointwise operations, is a commutative unital algebra.

**Definition 4.8** ($*$-Algebra)**.** *A $*$-algebra is an algebra $\mathcal{A}$ together with a map $a \mapsto a^*$ called an involution such that for all $a, b \in \mathcal{A}$ and $\alpha, \beta \in \mathbb{C}$,*

1. $a^{**} = a$*,*
2. $(ab)^* = b^* a^*$*,*
3. $(\alpha a + \beta b)^* = \overline{\alpha} a^* + \overline{\beta} b^*$*.*

**Example 4.12.** On $M_n(\mathbb{C})$ the involution is the conjugate transpose $A^* = \overline{A}^T$. It reverses products, $(AB)^* = B^* A^*$, and is conjugate-linear, so $M_n(\mathbb{C})$ is a $*$-algebra. On $C(X)$ the involution is pointwise complex conjugation $f^*(x) = \overline{f(x)}$.

**Definition 4.9** (Banach Algebra)**.** *A Banach algebra is an associative algebra $\mathcal{A}$ over $\mathbb{C}$ that is also a Banach space whose norm satisfies $\|xy\| \leq \|x\| \, \|y\|$ for all $x, y \in \mathcal{A}$.*

**Example 4.13.** $C(X)$ with the supremum norm is a Banach algebra, since $\|fg\|_\infty \leq \|f\|_\infty \|g\|_\infty$ and $C(X)$ is complete. The matrix algebra $M_n(\mathbb{C})$ with the operator norm is another example.

**Definition 4.10** ($C^*$-Algebra)**.** *A $C^*$-algebra is a Banach algebra $\mathcal{A}$ over $\mathbb{C}$ equipped with an involution $a \mapsto a^*$ (so that $\mathcal{A}$ is a Banach $*$-algebra) satisfying the $C^*$-identity*

$$\|a^* a\| = \|a\|^2 \qquad \text{for all } a \in \mathcal{A}. \tag{45}$$

The $C^*$-identity is the single axiom distinguishing $C^*$-algebras among Banach $*$-algebras, and it already forces the involution to be isometric, $\|a^*\| = \|a\|$. Indeed $\|a\|^2 = \|a^* a\| \leq \|a^*\| \, \|a\|$ gives $\|a\| \leq \|a^*\|$, and applying this to $a^*$ gives the reverse. Standard references are Murphy (19) and Kadison and Ringrose (15).

**Example 4.14.** The algebra $B(\mathcal{H})$ of bounded operators on a Hilbert space $\mathcal{H}$ is a $C^*$-algebra under the operator norm and the adjoint, and the $C^*$-identity $\|T^*T\| = \|T\|^2$ holds for every $T$. In finite dimensions this is $M_n(\mathbb{C})$, for instance for $T = \left(\begin{smallmatrix} 0 & 2 \\ 0 & 0 \end{smallmatrix}\right)$ one has $\|T\| = 2$ and $\|T^*T\| = \|\left(\begin{smallmatrix} 0 & 0 \\ 0 & 4 \end{smallmatrix}\right)\| = 4 = \|T\|^2$.

**Definition 4.11** (Adjoint Operator). *Let $\mathcal{H}$ be a Hilbert space and $T \in B(\mathcal{H})$. The adjoint $T^*$ is the unique bounded operator satisfying $\langle Tx, y\rangle = \langle x, T^*y\rangle$ for all $x, y \in \mathcal{H}$.*

If $T$ is represented by a matrix, then the adjoint is the conjugate transpose $T^* = \overline{T}^T$.

**Example 4.15.** For $T = \left(\begin{smallmatrix} 1 & i \\ 0 & 2 \end{smallmatrix}\right)$ on $\mathbb{C}^2$, the adjoint is $T^* = \left(\begin{smallmatrix} 1 & 0 \\ -i & 2 \end{smallmatrix}\right)$. One checks $\langle Tx, y\rangle = \langle x, T^*y\rangle$ for the standard inner product, which is the defining property of the adjoint.

**Definition 4.12** (von Neumann Algebra). *Let $\mathcal{H}$ be a Hilbert space. A von Neumann algebra is a $*$-subalgebra $M \subseteq B(\mathcal{H})$ such that*

1. $1_{\mathcal{H}} \in M$,
2. $A^* \in M$ *for all* $A \in M$,
3. $M$ *is closed in the weak operator topology.*

This is the concrete object. A $W^*$-algebra is the abstract version, a $C^*$-algebra that possesses a predual. By Sakai's theorem the two notions coincide, so we use the terms interchangeably (25, 27). Von Neumann algebras are the natural setting for non-commutative optimal transport, because the marginals of a coupling are naturally normal states, that is, elements of the predual.

**Example 4.16.** The full algebra $B(\mathcal{H})$ is a von Neumann algebra. Inside $M_2(\mathbb{C})$, the diagonal matrices $\{\mathrm{diag}(a, b) : a, b \in \mathbb{C}\}$ form a commutative von Neumann subalgebra, it is the algebra of functions on a two-point space, and it is exactly the diagonal subalgebra through which classical probability sits inside quantum probability.

## 4.5 Density Matrices

**Definition 4.13** (Density Operator). *Suppose a quantum state is represented by a unit vector $\psi \in \mathbb{C}^n$. The density operator is the rank-one projection $\rho = \psi\psi^*$, where $\psi^*$ is the conjugate transpose of $\psi$.*

More generally, let $\{\psi_\alpha\}$ be unit vectors and $\{p_\alpha\}$ a probability distribution, so $p_\alpha \geq 0$ and $\sum_\alpha p_\alpha = 1$. The density operator $\rho = \sum_\alpha p_\alpha \psi_\alpha \psi_\alpha^*$ describes a system prepared in state $\psi_\alpha$ with probability $p_\alpha$.

**Example 4.17.** For the qubit basis vector $\psi = \left(\begin{smallmatrix} 1 \\ 0 \end{smallmatrix}\right)$, the density operator is $\rho = \psi\psi^* = \left(\begin{smallmatrix} 1 & 0 \\ 0 & 0 \end{smallmatrix}\right)$, the projection onto the first coordinate. It records the definite state "spin up".

**Definition 4.14** (Density Matrix). *A quantum state on $\mathbb{C}^n$ is a matrix $\rho \in M_n(\mathbb{C})$ such that*

1. $\rho = \rho^\dagger$,
2. $\rho \geq 0$,
3. *Tr*$(\rho) = 1$.

A density matrix is the non-commutative analogue of a probability distribution: the map $a \mapsto \mathrm{Tr}(\rho\, a)$ is a state on $M_n(\mathbb{C})$, positive and normalised just as an expectation is.

**Example 4.18** (The maximally mixed qubit). The matrix $\rho = \frac{1}{2}I = \begin{pmatrix} \frac{1}{2} & 0 \\ 0 & \frac{1}{2} \end{pmatrix}$ is self-adjoint, positive, and has trace 1, so it is a density matrix. It represents complete uncertainty between the two basis states, the quantum counterpart of a fair coin. Its diagonal $(\frac{1}{2}, \frac{1}{2})$ is exactly the fair two-point probability vector.

**Definition 4.15** (Pure State). *A state $\rho \in M_n(\mathbb{C})$ is pure if $\rho = \psi\psi^*$ for a unit vector $\psi$, or equivalently* $\operatorname{rank}(\rho) = 1$, *or* $\rho^2 = \rho$.

**Example 4.19.** $\rho = \left(\begin{smallmatrix} 1 & 0 \\ 0 & 0 \end{smallmatrix}\right)$ is pure, since $\rho^2 = \rho$ and it has rank one. It corresponds to the definite state $\psi = \left(\begin{smallmatrix} 1 \\ 0 \end{smallmatrix}\right)$.

**Definition 4.16** (Mixed State). *A mixed state is a convex combination of pure states, $\rho = \sum_i p_i \psi_i \psi_i^*$ with $p_i > 0$ and $\sum_i p_i = 1$.*

**Example 4.20.** The maximally mixed qubit $\frac{1}{2}I = \frac{1}{2}\left(\begin{smallmatrix} 1 \\ 0 \end{smallmatrix}\right)\left(\begin{smallmatrix} 1 & 0 \end{smallmatrix}\right) + \frac{1}{2}\left(\begin{smallmatrix} 0 \\ 1 \end{smallmatrix}\right)\left(\begin{smallmatrix} 0 & 1 \end{smallmatrix}\right)$ is a mixed state: an equal mixture of spin up and spin down. It satisfies $\rho^2 = \frac{1}{4}I \neq \rho$, confirming it is not pure.

**Definition 4.17** (Faithful State). *A state $\rho$ is faithful if $\rho > 0$ (strictly positive definite), equivalently $\psi^*\rho\psi > 0$ for all non-zero $\psi \in \mathbb{C}^n$.*

**Example 4.21.** The maximally mixed state $\frac{1}{2}I$ is faithful, since all its eigenvalues equal $\frac{1}{2} > 0$. The pure state $\left(\begin{smallmatrix} 1 & 0 \\ 0 & 0 \end{smallmatrix}\right)$ is not faithful, because $\psi = \left(\begin{smallmatrix} 0 \\ 1 \end{smallmatrix}\right)$ gives $\psi^*\rho\psi = 0$. Faithful states assign positive weight to every direction and play the role of everywhere-positive densities.

## 4.6 Towards Non-Commutative Optimal Transport

The pieces are now in place to state the objects that the classical theory generalises. A density matrix $\rho \in M_n(\mathbb{C})$ is the non-commutative counterpart of a probability distribution, the commutative case being recovered by restricting to diagonal $\rho$. To form couplings we need marginals, provided by the partial trace.

**Definition 4.18** (Partial Trace). *Let $\mathcal{H}_A$ and $\mathcal{H}_B$ be finite-dimensional Hilbert spaces. The partial trace over $B$ is the unique linear map $\operatorname{Tr}_B : M(\mathcal{H}_A \otimes \mathcal{H}_B) \to M(\mathcal{H}_A)$ satisfying $\operatorname{Tr}_B(a \otimes b) = \operatorname{Tr}(b)\, a$ on elementary tensors, extended linearly. It is the non-commutative analogue of taking a marginal distribution.*

**Example 4.22.** For a product state $\rho \otimes \sigma$ on $\mathbb{C}^2 \otimes \mathbb{C}^2$, the partial trace over $B$ returns $\operatorname{Tr}_B(\rho \otimes \sigma) = \operatorname{Tr}(\sigma)\, \rho = \rho$, since $\operatorname{Tr}(\sigma) = 1$. Tracing out the second system recovers the state of the first, exactly as summing a joint distribution over one variable returns the marginal of the other.

**Definition 4.19** (Quantum Coupling). *Let $\rho \in M(\mathcal{H}_A)$ and $\sigma \in M(\mathcal{H}_B)$ be density matrices. A quantum coupling of $\rho$ and $\sigma$ is a density matrix $\omega \in M(\mathcal{H}_A \otimes \mathcal{H}_B)$ whose partial traces reproduce the marginals, $\operatorname{Tr}_B(\omega) = \rho$ and $\operatorname{Tr}_A(\omega) = \sigma$. Equivalently, as a state, $\omega(a \otimes 1) = \operatorname{Tr}(\rho\, a)$ and $\omega(1 \otimes b) = \operatorname{Tr}(\sigma\, b)$. Write $\Pi(\rho, \sigma)$ for the set of all such couplings. The product state $\rho \otimes \sigma$ always lies in $\Pi(\rho, \sigma)$, so this set is non-empty and convex.*

**Example 4.23.** For $\rho = \sigma = \frac{1}{2}I$ on $\mathbb{C}^2$, the product state $\omega = \frac{1}{2}I \otimes \frac{1}{2}I = \frac{1}{4}I_4$ is a quantum coupling: tracing out either factor returns $\frac{1}{2}I$. This is the quantum analogue of the independent coupling $\mu \otimes \nu$. Other, non-product couplings with the same marginals encode correlations (including entanglement) that have no classical counterpart.

Given a self-adjoint cost operator $C \in M(\mathcal{H}_A \otimes \mathcal{H}_B)$, one is led to the non-commutative Kantorovich problem $\min_{\omega \in \Pi(\rho,\sigma)} \operatorname{Tr}(\omega\, C)$, a semidefinite program whose dual mirrors Theorem 4.1, and to associated quantum Wasserstein distances.

*Remark:.* The definitions above are standard in the literature on quantum optimal transport (7, 9, 11). They serve as a bridge: Section 5 takes them up in earnest, first building the operator-algebraic foundation (states on $C^*$-algebras, completely positive maps and channels, the GNS construction, and tensor products of algebras) and then developing the non-commutative Monge and Kantorovich problems, a semidefinite Kantorovich duality, and the principal quantum Wasserstein distances.

# 5 Non-Commutative Optimal Transport

This section develops non-commutative optimal transport, the subject named in the title. The guiding principle is a dictionary between two theories. On the classical side, a probability measure lives on a space $X$, observables are functions in the commutative algebra $C(X)$, and a coupling is a joint measure. On the non-commutative side, the commutative algebra is replaced by a general $C^*$- or $W^*$-algebra $\mathcal{A}$, a probability measure is replaced by a state on $\mathcal{A}$, and a coupling is replaced by a state on a tensor product algebra with prescribed marginals. The following table summarises the correspondence.

| **Classical optimal transport** | **Non-commutative optimal transport** |
|---|---|
| sample space $X$ | unital $C^*$-algebra $\mathcal{A}$ |
| observable $f \in C(X)$ | self-adjoint element $a \in \mathcal{A}$ |
| probability measure $\mu$ | state $\varphi$ on $\mathcal{A}$ |
| expectation $\int f\,d\mu$ | value $\varphi(a)$ |
| marginal of a joint measure | partial trace / restriction of a state |
| coupling $\pi \in \Pi(\mu, \nu)$ | coupling state $\omega \in \Pi(\varphi, \psi)$ on $\mathcal{A} \otimes \mathcal{B}$ |
| cost function $c(x, y)$ | cost element $c = c^* \in \mathcal{A} \otimes \mathcal{B}$ |
| Kantorovich value $\inf_\pi \int c\,d\pi$ | $\inf_\omega \omega(c)$ |

We first assemble the operator-algebraic foundation that this dictionary requires, going beyond the density-matrix language of Section 4 to states on general algebras, completely positive maps, the GNS construction, and tensor products. We then develop the transport theory itself. Throughout, $\mathcal{A}$ and $\mathcal{B}$ denote unital $C^*$-algebras, and in the worked examples they are the finite-dimensional matrix algebras $M_n(\mathbb{C})$, where every analytic subtlety disappears and the optimisation problems become semidefinite programs. General references are Bratteli and Robinson (6), Murphy (19) and Paulsen (21) for operator algebras and completely positive maps, Nielsen and Chuang (20) and Wilde (30) for quantum channels, and De Palma et al. (9), Carlen and Maas (7) and Duvenhage (11) for the transport theory.

## 5.1 States on $C^*$-Algebras and Non-Commutative Probability Spaces

The density matrices of Section 4 are the states of the matrix algebra. We now record the general notion, which is the non-commutative replacement for a probability measure.

**Definition 5.1** (Positive Element). *An element $a$ of a $C^*$-algebra $\mathcal{A}$ is positive, written $a \geq 0$, if $a = a^*$ and its spectrum satisfies $\sigma(a) \subseteq [0, \infty)$. Equivalently, $a = b^*b$ for some $b \in \mathcal{A}$.*

**Example 5.1.** In $M_2(\mathbb{C})$ the matrix $a = \left(\begin{smallmatrix} 2 & 0 \\ 0 & 3 \end{smallmatrix}\right)$ is positive, since it is self-adjoint with eigenvalues $2, 3 > 0$, and indeed $a = b^*b$ with $b = \begin{pmatrix} \sqrt{2} & 0 \\ 0 & \sqrt{3} \end{pmatrix}$. In $C(X)$ the positive elements are exactly the functions $f \geq 0$.

**Definition 5.2** (State on a $C^*$-Algebra). *A state on a unital $C^*$-algebra $\mathcal{A}$ is a linear functional $\varphi : \mathcal{A} \to \mathbb{C}$ that is positive, meaning $\varphi(a) \geq 0$ whenever $a \geq 0$, and normalised, meaning $\varphi(1) = 1$. The set of all states on $\mathcal{A}$ is denoted $S(\mathcal{A})$, it is convex and, for a unital $C^*$-algebra, weak-$*$ compact.*

**Example 5.2** (Density matrices are states). On $\mathcal{A} = M_n(\mathbb{C})$, every density matrix $\rho$ defines a state $\varphi_\rho(a) = \mathrm{Tr}(\rho\, a)$, and conversely every state arises this way. For the qubit and $\rho = \frac{1}{2}I$, the state $\varphi_\rho(a) = \frac{1}{2}\mathrm{Tr}(a)$ returns the average of the two diagonal entries of $a$. On $C(X)$, a state is exactly a probability measure through $\varphi(f) = \int f\, d\mu$, by the Riesz–Markov theorem, which is why states are the correct non-commutative measures.

**Definition 5.3** (Non-Commutative Probability Space). *A non-commutative (or quantum) probability space is a pair $(\mathcal{A}, \varphi)$ consisting of a unital $C^*$-algebra $\mathcal{A}$ together with a state $\varphi$ on $\mathcal{A}$. It is called tracial if $\varphi(ab) = \varphi(ba)$ for all $a, b$, and faithful if $\varphi(a^*a) = 0$ implies $a = 0$.*

**Example 5.3.** The pair $(M_n(\mathbb{C}), \frac{1}{n}\mathrm{Tr})$ is a non-commutative probability space, the state $\tau(a) = \frac{1}{n}\mathrm{Tr}(a)$ is both tracial, since the trace is cyclic, and faithful, since $\mathrm{Tr}(a^*a) = 0$ forces $a = 0$. It is the non-commutative analogue of the uniform distribution on $n$ points, and reduces to it on the diagonal subalgebra.

### 5.2 Positive Maps, Completely Positive Maps and Quantum Channels

Transport moves one state to another, so we need the maps between algebras that carry states to states. The correct class is that of completely positive maps, which we now introduce.

**Definition 5.4** (Positive and Completely Positive Maps). *A linear map $\Phi : \mathcal{A} \to \mathcal{B}$ is positive if $\Phi(a) \geq 0$ whenever $a \geq 0$. It is completely positive if for every $n \in \mathbb{N}$ the amplified map $\Phi \otimes \mathrm{id}_n : \mathcal{A} \otimes M_n(\mathbb{C}) \to \mathcal{B} \otimes M_n(\mathbb{C})$ is positive.*

**Example 5.4** (Positive but not completely positive). The transpose $\Phi(a) = a^T$ on $M_2(\mathbb{C})$ is positive, since transposition preserves eigenvalues. It is not completely positive: applying $\Phi \otimes \mathrm{id}_2$ to the density matrix of the Bell state $\frac{1}{\sqrt{2}}(|00\rangle + |11\rangle)$ produces a matrix with a negative eigenvalue $-\frac{1}{2}$. This failure is exactly why complete positivity, and not mere positivity, is the physically correct requirement.

**Example 5.5** (A completely positive map). For any fixed $V \in M_{m\times n}(\mathbb{C})$, the map $\Phi(a) = VaV^*$ from $M_n(\mathbb{C})$ to $M_m(\mathbb{C})$ is completely positive, because $\Phi \otimes \mathrm{id}_k$ has the same form $X \mapsto (V \otimes I)X(V \otimes I)^*$ and conjugation preserves positivity.

**Theorem 5.1** (Stinespring Dilation, finite-dimensional Kraus form). *A linear map $\Phi : M_n(\mathbb{C}) \to M_m(\mathbb{C})$ is completely positive if and only if there exist matrices $E_1, \ldots, E_r \in M_{m\times n}(\mathbb{C})$, called Kraus operators, such that $\Phi(a) = \sum_{k=1}^{r} E_k\, a\, E_k^*$.*

**Example 5.6.** The depolarising map $\Phi(a) = \frac{1}{2}\mathrm{Tr}(a)\, I$ on $M_2(\mathbb{C})$ has the Kraus form with $E_k = \frac{1}{2}\sigma_k$ over the identity and the three Pauli matrices, giving $\Phi(a) = \frac{1}{4}\sum_k \sigma_k\, a\, \sigma_k = \frac{1}{2}\mathrm{Tr}(a)I$. It is therefore completely positive.

**Definition 5.5** (Quantum Channel). *A quantum channel is a completely positive, trace-preserving linear map $\Phi : M_n(\mathbb{C}) \to M_m(\mathbb{C})$, that is, a completely positive map with $\mathrm{Tr}(\Phi(a)) = \mathrm{Tr}(a)$ for all $a$. Equivalently, in Kraus form $\Phi(a) = \sum_k E_k a E_k^*$ with $\sum_k E_k^* E_k = I$. A channel maps density matrices to density matrices.*

**Example 5.7.** The partial trace $\mathrm{Tr}_B : M_n(\mathbb{C}) \otimes M_m(\mathbb{C}) \to M_n(\mathbb{C})$ of Section 4 is a quantum channel, with Kraus operators $E_k = I \otimes \langle k|$ for an orthonormal basis $\{|k\rangle\}$ of $\mathbb{C}^m$. It is the non-commutative analogue of forming a marginal distribution.

### 5.3 The Gelfand–Naimark–Segal Construction

The GNS construction realises any state as a vector state in a concrete Hilbert space. It is the tool that lets us treat abstract states geometrically and, in particular, it produces the non-commutative $L^2$-space on which the dynamical Wasserstein distance is defined.

**Theorem 5.2** (GNS Construction)**.** *Let $\mathcal{A}$ be a unital $C^*$-algebra and $\varphi$ a state on $\mathcal{A}$. Then there exist a Hilbert space $\mathcal{H}_\varphi$, a representation $\pi_\varphi : \mathcal{A} \to B(\mathcal{H}_\varphi)$ by bounded operators, and a cyclic unit vector $\Omega_\varphi \in \mathcal{H}_\varphi$ such that*

$$\varphi(a) = \langle \Omega_\varphi, \pi_\varphi(a)\, \Omega_\varphi \rangle \qquad \text{for all } a \in \mathcal{A}. \tag{46}$$

*The triple $(\mathcal{H}_\varphi, \pi_\varphi, \Omega_\varphi)$ is unique up to unitary equivalence.*

**Example 5.8** (GNS for a faithful matrix state)**.** Let $\mathcal{A} = M_n(\mathbb{C})$ and $\varphi(a) = \mathrm{Tr}(\rho\, a)$ with $\rho > 0$ faithful. The sesquilinear form $\langle a, b\rangle_\varphi = \varphi(a^*b) = \mathrm{Tr}(\rho\, a^*b)$ is a genuine inner product because $\rho$ is faithful, so $\mathcal{H}_\varphi = M_n(\mathbb{C})$ with this inner product, the representation is left multiplication $\pi_\varphi(a)b = ab$, and the cyclic vector is $\Omega_\varphi = I$. Then $\langle \Omega_\varphi, \pi_\varphi(a)\Omega_\varphi\rangle = \mathrm{Tr}(\rho\, a) = \varphi(a)$, as required. This Hilbert space is the non-commutative $L^2(\mathcal{A}, \varphi)$.

### 5.4 Tensor Products of Algebras and Couplings of States

Couplings live on tensor products of algebras. For the finite-dimensional algebras used in our examples the construction is elementary.

**Definition 5.6** (Tensor Product of Matrix Algebras)**.** *The tensor product of $M_n(\mathbb{C})$ and $M_m(\mathbb{C})$ is $M_n(\mathbb{C}) \otimes M_m(\mathbb{C}) \cong M_{nm}(\mathbb{C})$, with the product of elementary tensors given by the Kronecker product $(a \otimes b)(a' \otimes b') = (aa') \otimes (bb')$ and involution $(a \otimes b)^* = a^* \otimes b^*$. It is again a $C^*$-algebra.*

**Example 5.9.** $M_2(\mathbb{C}) \otimes M_2(\mathbb{C}) \cong M_4(\mathbb{C})$ is the algebra of operators on two qubits. The element $Z \otimes I$ is the $4 \times 4$ diagonal matrix $\mathrm{diag}(1, 1, -1, -1)$, and $I \otimes Z = \mathrm{diag}(1, -1, 1, -1)$, both self-adjoint elements of the joint algebra.

**Definition 5.7** (Coupling of States)**.** *Let $\varphi$ be a state on $\mathcal{A}$ and $\psi$ a state on $\mathcal{B}$. A coupling of $\varphi$ and $\psi$ is a state $\omega$ on $\mathcal{A} \otimes \mathcal{B}$ whose marginals are $\varphi$ and $\psi$, that is,*

$$\omega(a \otimes 1) = \varphi(a) \qquad \text{and} \qquad \omega(1 \otimes b) = \psi(b) \tag{47}$$

*for all $a \in \mathcal{A}$, $b \in \mathcal{B}$. In the matrix case, writing $\omega(x) = \mathrm{Tr}(\Omega\, x)$ for a density matrix $\Omega$ on $\mathcal{H}_A \otimes \mathcal{H}_B$, this is $\mathrm{Tr}_B(\Omega) = \rho$ and $\mathrm{Tr}_A(\Omega) = \sigma$. The set of all couplings is written $\Pi(\varphi, \psi)$.*

**Example 5.10** (The product coupling and a correlated coupling)**.** For $\varphi = \psi = \frac{1}{2}\mathrm{Tr}$ on $M_2(\mathbb{C})$ (the state of $\rho = \sigma = \frac{1}{2}I$), the product state $\Omega = \frac{1}{2}I \otimes \frac{1}{2}I = \frac{1}{4}I_4$ is a coupling, the analogue of the independent joint distribution. A different coupling with the same marginals is $\Omega' = \frac{1}{2}(|00\rangle\langle 00| + |11\rangle\langle 11|)$, which correlates the two qubits while still tracing down to $\frac{1}{2}I$ on each side. The maximally entangled coupling $|\Phi^+\rangle\langle\Phi^+|$ has the same marginals again, and encodes a correlation with no classical analogue.

**Proposition 5.1** (Basic properties of $\Pi(\varphi,\psi)$)**.** *For states $\varphi$ on $\mathcal{A}$ and $\psi$ on $\mathcal{B}$, the set $\Pi(\varphi,\psi)$ is non-empty, convex and, in the finite-dimensional case, compact.*

*Proof.* The product state $\varphi\otimes\psi$, defined on elementary tensors by $(\varphi\otimes\psi)(a\otimes b) = \varphi(a)\psi(b)$, is a state with the required marginals, so $\Pi(\varphi,\psi) \neq \emptyset$. If $\omega_1, \omega_2 \in \Pi(\varphi,\psi)$ and $t \in [0,1]$, then $t\omega_1 + (1-t)\omega_2$ is again a state whose marginals are the same convex combination of $\varphi$ and $\psi$, namely $\varphi$ and $\psi$, hence $\Pi(\varphi,\psi)$ is convex. In finite dimensions the marginal constraints are linear equalities on the compact convex set of density matrices, so $\Pi(\varphi,\psi)$ is compact. $\square$

**Example 5.11.** For $\varphi = \psi = \frac{1}{2}\operatorname{Tr}$ the three couplings $\frac{1}{4}I_4$, $\frac{1}{2}(|00\rangle\langle 00| + |11\rangle\langle 11|)$ and $|\Phi^+\rangle\langle\Phi^+|$ above are corners and interior points of the same convex, compact set $\Pi(\varphi,\psi)$, illustrating the proposition.

**Proposition 5.2** (Recovery of classical couplings)**.** *Let $\mathcal{A} = \mathcal{B} = \mathbb{C}^n$, viewed as the diagonal subalgebra of $M_n(\mathbb{C})$, so that states are probability vectors $p, q$ on $\{1,\dots,n\}$. Then the couplings in $\Pi(p,q)$ that are supported on the diagonal subalgebra $\mathbb{C}^n \otimes \mathbb{C}^n$ are exactly the classical couplings of $p$ and $q$, that is, the joint distributions with marginals $p$ and $q$.*

*Proof.* A state on the commutative algebra $\mathbb{C}^n \otimes \mathbb{C}^n \cong \mathbb{C}^{n\times n}$ is a probability vector $(\pi_{ij})$, and the marginal conditions $\omega(a\otimes 1) = p(a)$ and $\omega(1\otimes b) = q(b)$ read $\sum_j \pi_{ij} = p_i$ and $\sum_i \pi_{ij} = q_j$, which are precisely the classical marginal constraints. $\square$

**Example 5.12.** With $p = (0.6, 0.4)$ and $q = (0.5, 0.5)$ the classical coupling $\pi = \left(\begin{smallmatrix} 0.3 & 0.3 \\ 0.2 & 0.2 \end{smallmatrix}\right)$ of Section 4 sits inside $\Pi(p,q)$ as the diagonal density matrix $\operatorname{diag}(0.3, 0.3, 0.2, 0.2)$ on $\mathbb{C}^4$. The quantum theory strictly enlarges the classical one by also allowing non-diagonal couplings.

### 5.5 The Non-Commutative Monge and Kantorovich Problems

**Definition 5.8** (Cost Element and the Non-Commutative Kantorovich Problem)**.** *A cost element is a self-adjoint $c = c^* \in \mathcal{A}\otimes\mathcal{B}$, interpreted as the observable whose expectation is the transport cost. The non-commutative Kantorovich problem associated with states $\varphi, \psi$ and cost $c$ is*

$$\mathbb{K}(\varphi,\psi,c) = \inf_{\omega\in\Pi(\varphi,\psi)} \omega(c). \tag{48}$$

*In the matrix case, with $c$ represented by a Hermitian matrix $C$, this is $\inf_\Omega \operatorname{Tr}(\Omega\, C)$ over couplings $\Omega$, a semidefinite program.*

**Example 5.13** (A qubit transport cost)**.** On two qubits take the Hermitian cost $C = (Z\otimes I - I\otimes Z)^2$. Using $Z\otimes I = \operatorname{diag}(1,1,-1,-1)$ and $I\otimes Z = \operatorname{diag}(1,-1,1,-1)$ gives $Z\otimes I - I\otimes Z = \operatorname{diag}(0,2,-2,0)$ and hence $C = \operatorname{diag}(0,4,4,0)$. This cost charges 4 for the two configurations in which the qubits disagree and 0 when they agree, the quantum version of the discrete cost $c(x,y) = (x-y)^2$ on $\{+1,-1\}$.

*Remark:* (The Monge analogue). The classical Monge problem optimises over deterministic couplings, those concentrated on the graph of a map. Its non-commutative analogue would optimise over couplings induced by a single $*$-homomorphism or channel intertwining the two states. Such deterministic couplings are delicate in the non-commutative setting and need not exist for given marginals, in sharp contrast to the classical case where they exist generically. This is one more reason, beyond the ones seen classically in Section 4, why the Kantorovich formulation over all couplings is the correct general setting.

## 5.6 Non-Commutative Kantorovich Duality

The classical Kantorovich problem has a dual over pairs of potentials, Theorem 4.1. The non-commutative problem, being a semidefinite program in finite dimensions, has a matching dual over pairs of self-adjoint elements. This is one of the two results of this section that we prove in full.

**Theorem 5.3** (Non-Commutative Kantorovich Duality)**.** *Let $\mathcal{A} = M_n(\mathbb{C})$ and $\mathcal{B} = M_m(\mathbb{C})$ with states $\varphi(a) = \mathrm{Tr}(\rho\, a)$ and $\psi(b) = \mathrm{Tr}(\sigma\, b)$, and let $C = C^* \in \mathcal{A} \otimes \mathcal{B}$ be a cost element. Then*

$$\inf_{\omega \in \Pi(\varphi,\psi)} \omega(C) = \sup_{\substack{K=K^* \in \mathcal{A},\, L=L^* \in \mathcal{B} \\ K \otimes I + I \otimes L \leq C}} \Big( \mathrm{Tr}(\rho\, K) + \mathrm{Tr}(\sigma\, L) \Big), \tag{49}$$

*and both the infimum and the supremum are attained.*

*Proof.* Write the primal as $\min_\Omega \mathrm{Tr}(\Omega C)$ subject to $\Omega \geq 0$, $\mathrm{Tr}_B \Omega = \rho$, $\mathrm{Tr}_A \Omega = \sigma$, the constraint $\mathrm{Tr}\, \Omega = 1$ is implied by the marginal constraints. Introduce Hermitian multipliers $K$ on $\mathcal{H}_A$ and $L$ on $\mathcal{H}_B$ for the two marginal constraints and form the Lagrangian

$$\mathcal{L}(\Omega, K, L) = \mathrm{Tr}(\Omega C) - \mathrm{Tr}\big(K(\mathrm{Tr}_B \Omega - \rho)\big) - \mathrm{Tr}\big(L(\mathrm{Tr}_A \Omega - \sigma)\big). \tag{50}$$

Using the identity $\mathrm{Tr}(K\, \mathrm{Tr}_B \Omega) = \mathrm{Tr}((K \otimes I)\Omega)$, and likewise for $L$, this becomes

$$\mathcal{L}(\Omega, K, L) = \mathrm{Tr}\big((C - K \otimes I - I \otimes L)\, \Omega\big) + \mathrm{Tr}(\rho K) + \mathrm{Tr}(\sigma L). \tag{51}$$

Minimising over $\Omega \geq 0$ gives $\mathrm{Tr}(\rho K) + \mathrm{Tr}(\sigma L)$ if $C - K \otimes I - I \otimes L \geq 0$, and $-\infty$ otherwise. The dual problem is therefore to maximise $\mathrm{Tr}(\rho K) + \mathrm{Tr}(\sigma L)$ subject to $K \otimes I + I \otimes L \leq C$. Weak duality holds by construction. Strong duality and attainment follow from the semidefinite-programming strong-duality theorem, whose Slater condition is met because the product coupling $\rho \otimes \sigma$ is a strictly feasible primal point. $\square$

**Example 5.14** (Primal equals dual for two classical states)**.** Take $\rho = |0\rangle\langle 0|$ and $\sigma = |1\rangle\langle 1|$ on the qubit, with the cost $C = \mathrm{diag}(0, 4, 4, 0)$ above. The only coupling of $\rho$ and $\sigma$ is the product $|0\rangle\langle 0| \otimes |1\rangle\langle 1| = |01\rangle\langle 01|$, so the primal value is $\mathrm{Tr}(C\, |01\rangle\langle 01|) = 4$. On the dual side, choosing the diagonal $K = \mathrm{diag}(k_0, k_1)$ and $L = \mathrm{diag}(l_0, l_1)$ with $k_0 = 4$, $l_1 = 0$, $l_0 = -4$, $k_1 = 0$ satisfies $K \otimes I + I \otimes L \leq C$ and gives $\mathrm{Tr}(\rho K) + \mathrm{Tr}(\sigma L) = k_0 + l_1 = 4$. Primal and dual agree, as Theorem 5.3 guarantees.

## 5.7 Recovery of Classical Optimal Transport

A non-commutative theory is only convincing if it contains the classical one. The next theorem, the second we prove in full, shows that on diagonal data the non-commutative Kantorovich value coincides with the classical one, so classical optimal transport is exactly the commutative special case.

**Theorem 5.4** (Classical recovery)**.** *Let $\rho = \mathrm{diag}(p)$ and $\sigma = \mathrm{diag}(q)$ be diagonal density matrices on $\mathbb{C}^n$ and $\mathbb{C}^m$, corresponding to probability vectors $p$ and $q$, and let the cost element be diagonal, $C = \mathrm{diag}\big(c(i,j)\big)$ on $\mathbb{C}^n \otimes \mathbb{C}^m$. Then*

$$\inf_{\omega \in \Pi(\varphi,\psi)} \omega(C) = \min_{\pi \in \Pi(p,q)} \sum_{i,j} c(i,j)\, \pi_{ij}, \tag{52}$$

*the right-hand side being the classical Kantorovich value with cost $c$.*

*Proof.* Let $\Omega \in \Pi(\varphi, \psi)$ be any quantum coupling and write $d = \mathrm{diag}(\Omega)$ for its diagonal, indexed by pairs $(i, j)$. Since $C$ is diagonal, $\mathrm{Tr}(\Omega C) = \sum_{i,j} c(i,j)\, d_{ij}$, so the objective depends on $\Omega$ only through $d$. The marginal constraints force $\sum_j d_{ij} = p_i$ and $\sum_i d_{ij} = q_j$, so $d$ is a classical coupling, giving $\mathrm{Tr}(\Omega C) \geq \min_{\pi \in \Pi(p,q)} \sum c(i,j)\pi_{ij}$. Conversely, any classical coupling $\pi \in \Pi(p, q)$ embeds as the diagonal quantum coupling $\Omega = \mathrm{diag}(\pi)$, which lies in $\Pi(\varphi, \psi)$ and attains $\mathrm{Tr}(\Omega C) = \sum c(i,j)\pi_{ij}$. Taking the infimum over both sides gives equality. □

**Example 5.15.** For $p = (1, 0)$, $q = (0, 1)$ and the diagonal cost $c(i, j) = (x_i - x_j)^2$ with $x_1 = +1$, $x_2 = -1$, so $c(1, 2) = 4$, the classical value is 4, achieved by the only coupling that moves all mass from state 1 to state 2. This matches the quantum value computed in the previous subsection, confirming that the non-commutative problem reduces to the classical one on diagonal data.

## 5.8 Quantum Wasserstein Distances

Specialising the cost element turns the non-commutative Kantorovich value into a distance between states. Several inequivalent constructions appear in the literature, adapted to different purposes. We present the three most influential, keeping the exposition at the level of definitions and stated properties, with attributions.

### 5.8.1 A static Wasserstein-2 distance from couplings

**Definition 5.9** (Static quantum $W_2$)**.** *Let $\mathcal{A} = \mathcal{B}$ and let $c = c^* \geq 0 \in \mathcal{A} \otimes \mathcal{A}$ be a cost element that plays the role of a squared distance, symmetric under the flip $a \otimes b \mapsto b \otimes a$. The static quantum Wasserstein-2 quantity between states $\varphi$ and $\psi$ is*

$$W_2(\varphi, \psi) = \Big( \inf_{\omega \in \Pi(\varphi,\psi)} \omega(c) \Big)^{1/2}. \tag{53}$$

This is the direct transcription of the classical formula $W_2(\mu, \nu)^2 = \inf_\pi \int d(x, y)^2\, d\pi$. Symmetry $W_2(\varphi, \psi) = W_2(\psi, \varphi)$ follows from the flip-symmetry of $c$. That $W_2$ separates states and satisfies the triangle inequality is not automatic in the non-commutative setting and requires structural hypotheses on $c$, establishing such metric properties is a central theme of the coupling approach developed by Duvenhage (11).

**Example 5.16.** On the qubit take $c = (Z \otimes I - I \otimes Z)^2 \geq 0$, which is flip-symmetric. For the two orthogonal states $\rho = |0\rangle \langle 0|$ and $\sigma = |1\rangle \langle 1|$ the infimum equals 4, so $W_2(\varphi, \psi) = 2$. For $\rho = \sigma$ the diagonal-agreeing coupling $\frac{1}{2}(|00\rangle \langle 00| + |11\rangle \langle 11|)$ gives cost 0, so $W_2(\varphi, \varphi) = 0$, as a distance should.

### 5.8.2 The quantum Wasserstein distance of order 1

De Palma, Marvian, Trevisan and Lloyd (9) defined a quantum $W_1$ distance on systems of $n$ qudits, $\mathcal{H} = (\mathbb{C}^d)^{\otimes n}$, that reduces to the classical $W_1$ for the Hamming distance on diagonal states. It is defined through a dual, Lipschitz-type description.

**Definition 5.10** (Quantum Lipschitz constant and quantum $W_1$)**.** *For a Hermitian observable $H$ on $\mathcal{H} = (\mathbb{C}^d)^{\otimes n}$, the quantum Lipschitz constant is*

$$\|H\|_L = 2 \max_{1 \leq i \leq n} \min_{H_i} \|H - H_i\|_\infty, \tag{54}$$

*where $H_i$ ranges over Hermitian operators that act trivially on the $i$-th qudit and $\|\cdot\|_\infty$ is the operator norm. The quantum Wasserstein distance of order* 1 *between states $\rho$ and $\sigma$ is*

$$W_1(\rho,\sigma) = \max_{\|H\|_L \le 1} \mathrm{Tr}\big(H(\rho-\sigma)\big). \tag{55}$$

**Example 5.17** (One qudit reduces to the trace distance)**.** For a single qudit, $n = 1$, the only subsystem is the whole system, and $\min_{c\in\mathbb{R}} \|H - cI\|_\infty = \frac{1}{2}(\lambda_{\max}(H) - \lambda_{\min}(H))$, so $\|H\|_L = \lambda_{\max}(H) - \lambda_{\min}(H)$ is the spectral spread. Maximising $\mathrm{Tr}(H(\rho-\sigma))$ over observables of spread at most 1 gives $W_1(\rho,\sigma) = \frac{1}{2}\|\rho-\sigma\|_1$, the trace distance. For $\rho = |0\rangle\langle 0|$ and $\sigma = |1\rangle\langle 1|$ this equals 1, matching the classical Hamming distance between the two distinguishable states.

**Theorem 5.5** (Classical reduction)**.** *If $\rho$ and $\sigma$ are diagonal in the computational basis, corresponding to probability distributions $p$ and $q$ on $\{1,\dots,d\}^n$, then $W_1(\rho,\sigma)$ equals the classical Wasserstein-*1 *distance between $p$ and $q$ for the Hamming distance on $\{1,\dots,d\}^n$ (9).*

**Example 5.18.** Two computational-basis states of $n$ qubits that differ in exactly $k$ positions, such as $|00\cdots 0\rangle$ and a string with $k$ ones, are at quantum $W_1$ distance $k$, the number of qudits that must be changed. This is the Hamming distance, so the quantum $W_1$ generalises the natural edit-distance cost.

### 5.8.3 The dynamical quantum Wasserstein-2 metric

Carlen and Maas (7) introduced a Riemannian, dynamical $W_2$ on the faithful states, by transcribing the Benamou–Brenier and Otto picture of Section 4 to the non-commutative setting. The construction fixes a faithful state $\sigma$ and a quantum Markov semigroup $P_t = e^{t\mathcal{L}}$ satisfying detailed balance with respect to $\sigma$, its generator $\mathcal{L}$ factors through a non-commutative gradient $\nabla$ built from the derivations that implement $\mathcal{L}$.

**Definition 5.11** (Dynamical quantum $W_2$, informal)**.** *With $\nabla$ the non-commutative gradient and a continuity equation $\dot\rho_t + \mathrm{div}(\rho_t\nabla\phi_t) = 0$ relating a curve of states $\rho_t$ to a curve of potentials $\phi_t$, the dynamical Wasserstein-*2 *distance is*

$$W_2(\rho_0,\rho_1) = \inf\left\{\Big(\int_0^1 \langle\nabla\phi_t, \rho_t\nabla\phi_t\rangle\,dt\Big)^{1/2} \;:\; \dot\rho_t + \mathrm{div}(\rho_t\nabla\phi_t) = 0,\ \rho_t\big|_{0,1} = \rho_0,\rho_1\right\}, \tag{56}$$

*the infimum being over curves of states joining $\rho_0$ to $\rho_1$.*

**Theorem 5.6** (Entropy as gradient flow)**.** *With respect to the metric $W_2$ above, the quantum Markov semigroup $P_t$ is the gradient flow of the relative entropy $D(\rho\,\|\,\sigma) = \mathrm{Tr}(\rho\log\rho - \rho\log\sigma)$, that is, the evolution $\dot\rho_t = \mathcal{L}^*\rho_t$ is the steepest descent of $D(\cdot\,\|\,\sigma)$ for the Riemannian structure defined by $W_2$ (7).*

**Example 5.19.** For a single qubit with the dephasing semigroup, whose action shrinks the off-diagonal entries of a density matrix towards zero while fixing the diagonal, the invariant state is $\sigma = \frac{1}{2}I$ and the relative entropy $D(\rho\,\|\,\frac{1}{2}I) = \log 2 - S(\rho)$ decreases along the evolution, where $S$ is the von Neumann entropy. The dephasing dynamics is the gradient flow that drives every state towards the maximally mixed state, the non-commutative analogue of heat flow relaxing a distribution to the uniform one.

## 5.9 Connections and Outlook

The non-commutative theory inherits, and in places sharpens, the connections of its classical parent.

**Entropic regularisation and quantum Schrödinger bridges.** Adding an entropic penalty to the Kantorovich functional, $\inf_{\omega\in\Pi(\varphi,\psi)}\big(\omega(c)+\varepsilon\, D(\omega\,\|\,\varphi\otimes\psi)\big)$ with $D$ the relative entropy of the coupling against the product state, yields a strictly convex problem whose classical analogue is the Schrödinger bridge (16, 22). In the matrix case this is a matrix-scaling problem solvable by a quantum analogue of the Sinkhorn iteration, and it interpolates between the transport cost as $\varepsilon\to 0$ and the product coupling as $\varepsilon\to\infty$.

**Distributional robustness and risk.** The classical link between Wasserstein distances and distributionally robust optimisation (18) suggests a non-commutative counterpart, in which a worst-case expectation $\sup\{\omega'(c) : W(\omega',\omega)\le\varepsilon\}$ is taken over a quantum Wasserstein ball of states. Whether this yields tractable robust bounds for quantum observables, and whether the actuarial reading of Section 4 (Wasserstein distance controlling risk measures built from quantile functions) has a genuine non-commutative form, is a natural and, at present, largely open question, we flag it as speculative rather than established.

**What remains.** The material of this section is an exposition of the established theory. Several foundational questions remain active. Chief among them are: general conditions under which a static quantum $W_p$ satisfies all metric axioms, beyond the finite-dimensional and diagonal cases, the exact relationship between the static coupling distances, the order-1 distance of (9) and the dynamical distance of (7), which are known to differ in general, and the extension of the duality Theorem 5.3 beyond finite dimensions, where the semidefinite-programming argument no longer applies and infinite-dimensional convex duality, with its attendant duality gaps, must be used instead. These are the directions in which original contributions to the theory naturally lie.

# 6 Applications in Actuarial Science and Risk Modelling

Optimal transport is not only a subject of pure mathematics, the Wasserstein distances and the Kantorovich problem have become working tools in quantitative risk management and actuarial science. The reason is structural. An insurance liability is a distribution of losses, a coupling is a model of dependence between risks, and a premium or a capital charge is a functional of the loss distribution. Optimal transport is exactly the calculus of distributions, dependence and their functionals, so it speaks directly to the questions actuaries ask: how far apart are two loss models, how sensitive is a capital figure to the choice of model, what is the worst dependence structure consistent with the data, and how should several models be combined. This section develops these applications, drawing on the classical theory of Sections 2 to 4. Standard references are McNeil, Frey and Embrechts (17), Rüschendorf (24) and Föllmer and Schied (14). Throughout, a loss is a real random variable $X$, with large positive values unfavourable, and $F_X$ denotes its cumulative distribution function with quantile (inverse) function $F_X^{-1}(u)=\inf\{x : F_X(x)\ge u\}$.

## 6.1 Distances Between Loss Models

The first use of optimal transport in this setting is as a metric on loss models, quantifying model risk. Two candidate severity distributions fitted to the same claims data can be compared by the Wasserstein distance between them, which by the one-dimensional formula of Section 4 has a transparent closed form.

**Definition 6.1** (Wasserstein distance between loss models)**.** *For loss random variables $X$ and $Y$ with finite p-th moments, the Wasserstein-p distance between their laws is*

$$W_p(X,Y) = \left( \int_0^1 \left| F_X^{-1}(u) - F_Y^{-1}(u) \right|^p du \right)^{1/p}, \tag{57}$$

*and in particular $W_1(X,Y) = \int_0^1 |F_X^{-1}(u) - F_Y^{-1}(u)|\, du = \int_{-\infty}^{\infty} |F_X(t) - F_Y(t)|\, dt$.*

**Example 6.1** (Two exponential severity models)**.** Suppose claim sizes are modelled either as $X \sim$ Exp with mean $\theta_1 = 1000$ or as $Y \sim$ Exp with mean $\theta_2 = 1200$, so $F_X^{-1}(u) = -\theta_1 \ln(1-u)$ and $F_Y^{-1}(u) = -\theta_2 \ln(1-u)$. Then

$$W_1(X,Y) = \int_0^1 |\theta_1 - \theta_2| \big( - \ln(1-u) \big)\, du = |\theta_1 - \theta_2| \int_0^1 (-\ln t)\, dt = |\theta_1 - \theta_2| = 200, \tag{58}$$

using $\int_0^1 (-\ln t)\, dt = 1$. For exponential models the Wasserstein-1 distance is simply the gap between the mean claim sizes, and it reads as the expected absolute discrepancy in modelled claim amount, a natural measure of model risk in the severity assumption.

### 6.2 Risk Measures and Their Robustness

Regulatory capital and premiums are computed through risk measures. The two standard choices are the Value-at-Risk and the Tail-Value-at-Risk (also called Expected Shortfall).

**Definition 6.2** (Value-at-Risk and Tail-Value-at-Risk)**.** *For a loss $X$ and a confidence level $\alpha \in (0,1)$, the Value-at-Risk is the quantile* $\mathrm{VaR}_\alpha(X) = F_X^{-1}(\alpha)$*, and the Tail-Value-at-Risk is the average of the quantile above that level,*

$$\mathrm{TVaR}_\alpha(X) = \frac{1}{1-\alpha} \int_\alpha^1 F_X^{-1}(u)\, du. \tag{59}$$

**Example 6.2.** For $X \sim$ Exp with mean $\theta$ one computes $\mathrm{VaR}_\alpha(X) = -\theta \ln(1-\alpha)$ and $\mathrm{TVaR}_\alpha(X) = \theta\,(1 - \ln(1-\alpha))$. At $\alpha = 0.99$ and $\theta = 1000$ this gives $\mathrm{VaR}_{0.99}(X) = 4605$ and $\mathrm{TVaR}_{0.99}(X) = 5605$, the latter being larger because it averages the entire tail beyond the 99% quantile rather than reading off a single point.

A basic question of model risk is how much a capital figure can move when the loss model is perturbed. Optimal transport answers this cleanly for Tail-Value-at-Risk, through the quantile representation, and warns against Value-at-Risk.

**Proposition 6.1** (Tail-Value-at-Risk is Wasserstein-Lipschitz)**.** *For all loss models $X, Y$ and every $\alpha \in (0,1)$,*

$$\big|\mathrm{TVaR}_\alpha(X) - \mathrm{TVaR}_\alpha(Y)\big| \le \frac{1}{1-\alpha}\, W_1(X,Y). \tag{60}$$

*Proof.* Using the quantile representation and the triangle inequality,

$$\big|\mathrm{TVaR}_\alpha(X) - \mathrm{TVaR}_\alpha(Y)\big| = \frac{1}{1-\alpha} \left| \int_\alpha^1 \big(F_X^{-1}(u) - F_Y^{-1}(u)\big)\, du \right| \le \frac{1}{1-\alpha} \int_\alpha^1 \big|F_X^{-1}(u) - F_Y^{-1}(u)\big|\, du. \tag{61}$$

Extending the integral from $[\alpha, 1]$ to $[0,1]$ only increases it, and the result equals $\frac{1}{1-\alpha} W_1(X,Y)$ by the quantile formula for $W_1$. $\square$

**Example 6.3.** The two exponential models above satisfy $W_1(X, Y) = 200$, so by Proposition 6.1 their Tail-Value-at-Risk at $\alpha = 0.99$ differ by at most $200/0.01 = 20000$. The actual difference is $\mathrm{TVaR}_{0.99}(Y) - \mathrm{TVaR}_{0.99}(X) = 1200 \cdot 5.605 - 1000 \cdot 5.605 = 1121$, comfortably inside the bound. The point of the proposition is qualitative: a small Wasserstein change in the loss model produces only a proportionally small change in the Tail-Value-at-Risk, so this capital measure is stable under model uncertainty.

**Example 6.4** (Value-at-Risk is not Wasserstein-continuous)**.** Contrast this with Value-at-Risk. Let $X = \frac{1}{2}\delta_0 + \frac{1}{2}\delta_2$ and $Y = (\frac{1}{2} - \eta)\delta_0 + (\frac{1}{2} + \eta)\delta_2$ for small $\eta > 0$. Transporting mass $\eta$ from 0 to 2 gives $W_1(X, Y) = 2\eta \to 0$ as $\eta \to 0$. However $F_X(0) = \frac{1}{2} \geq \frac{1}{2}$ gives $\mathrm{VaR}_{1/2}(X) = 0$, whereas $F_Y(0) = \frac{1}{2} - \eta < \frac{1}{2}$ gives $\mathrm{VaR}_{1/2}(Y) = 2$. Thus an arbitrarily small Wasserstein perturbation moves the Value-at-Risk by a fixed amount 2. Value-at-Risk is therefore discontinuous in the Wasserstein metric wherever the loss distribution is flat at the confidence level, which is one mathematical reason regulators have moved towards Tail-Value-at-Risk for capital.

### 6.3 Distributionally Robust Premiums and Reserves

When the loss model is estimated from finite data, one may hedge against estimation error by optimising against the worst model within a Wasserstein ball of the empirical distribution. This is Wasserstein distributionally robust optimisation (4, 18).

**Definition 6.3** (Wasserstein ambiguity set and robust expectation)**.** *Let $\widehat{P}$ be the empirical loss distribution from data and $\varepsilon \geq 0$ an ambiguity radius. The Wasserstein ambiguity set is $\mathcal{B}_\varepsilon(\widehat{P}) = \{Q : W_1(Q, \widehat{P}) \leq \varepsilon\}$, and for a loss functional $\ell$ the worst-case expected loss is*

$$\sup_{Q \in \mathcal{B}_\varepsilon(\widehat{P})} \mathbb{E}_Q[\ell(X)]. \tag{62}$$

**Theorem 6.1** (Strong duality for Wasserstein robustness)**.** *For upper semicontinuous $\ell$ with $|\ell(x) - \ell(y)|$ controlled by $|x - y|$, the worst-case expected loss admits the dual representation*

$$\sup_{Q \in \mathcal{B}_\varepsilon(\widehat{P})} \mathbb{E}_Q[\ell(X)] = \inf_{\lambda \geq 0} \left\{ \lambda\varepsilon + \mathbb{E}_{\widehat{P}}\Big[ \sup_x \big(\ell(x) - \lambda|x - X|\big) \Big] \right\}. \tag{63}$$

*In particular, if $\ell$ is 1-Lipschitz then the worst-case equals $\mathbb{E}_{\widehat{P}}[\ell(X)] + \varepsilon$.*

**Example 6.5** (A robust pure premium)**.** Take the pure premium functional $\ell(x) = x$, which is 1-Lipschitz. The robust pure premium over a Wasserstein ball of radius $\varepsilon$ is

$$\sup_{Q \in \mathcal{B}_\varepsilon(\widehat{P})} \mathbb{E}_Q[X] = \overline{X} + \varepsilon, \tag{64}$$

where $\overline{X}$ is the empirical mean loss. The transport radius $\varepsilon$ appears as an explicit safety loading on top of the expected loss, giving the classical expected-value premium principle a precise interpretation in terms of ambiguity aversion: the loading is the price of robustness against a Wasserstein-$\varepsilon$ misspecification of the loss distribution.

## 6.4 Dependence Uncertainty, Comonotonicity and Risk Aggregation

The deepest actuarial application of optimal transport concerns dependence. An insurer holds many risks whose marginal distributions are estimated with some confidence, but whose joint dependence is largely unknown. The set of admissible joint models with fixed marginals $F$ and $G$ is exactly the Fréchet class, which is the transport polytope $\Pi(F, G)$ of couplings from Section 4. Optimal transport therefore governs the range of possible aggregate risks.

**Definition 6.4** (Comonotonic coupling)**.** *Two losses $X$ and $Y$ are comonotone if $(X, Y)$ has the same law as $(F_X^{-1}(U), F_Y^{-1}(U))$ for a single uniform random variable $U$ on $[0, 1]$. The corresponding coupling is the comonotonic element of the Fréchet class $\Pi(F_X, F_Y)$, and it is the upper Fréchet–Hoeffding bound.*

**Example 6.6.** If $X$ and $Y$ are both Exp with mean 1000, their comonotonic coupling sets $X = Y = -1000 \ln(1 - U)$, so the two losses always move together and $Y = X$ almost surely. This is the strongest possible positive dependence, in which a large claim on one risk always accompanies a large claim on the other.

The comonotonic coupling is the optimal transport plan for the problem of maximising a supermodular reward, the monotone rearrangement of Section 4. Its actuarial significance is that it produces the worst aggregate risk for coherent capital measures.

**Proposition 6.2** (Comonotonicity is the worst case for Tail-Value-at-Risk)**.** *Let $S = X + Y$ where $X, Y$ have fixed marginals but arbitrary dependence. Then for every $\alpha$,*

$$\mathrm{TVaR}_\alpha(X + Y) \leq \mathrm{TVaR}_\alpha(X) + \mathrm{TVaR}_\alpha(Y) = \mathrm{TVaR}_\alpha(X^c + Y^c), \tag{65}$$

*where $(X^c, Y^c)$ is the comonotonic coupling. Hence the comonotonic dependence maximises the Tail-Value-at-Risk of the aggregate loss over the whole Fréchet class.*

*Proof sketch.* Tail-Value-at-Risk is subadditive, giving the inequality, and it is additive under comonotonicity, since for comonotone risks the quantile of the sum is the sum of the quantiles, giving the equality (10). Together these bound $\mathrm{TVaR}_\alpha(X + Y)$ by the comonotonic value. □

**Example 6.7.** For two Exp(1000) losses, $\mathrm{TVaR}_{0.99}$ of each is 5605, so the comonotonic aggregate has $\mathrm{TVaR}_{0.99}(X^c + Y^c) = 11210$. Any other dependence, for instance independence, gives a strictly smaller Tail-Value-at-Risk, because subadditivity is strict away from comonotonicity. The figure 11210 is thus the conservative capital charge that holds no matter what the true dependence is, which is exactly the quantity a prudent reserving standard requires.

*Remark:* (Worst-case Value-at-Risk is harder). The clean picture of Proposition 6.2 relies on the subadditivity and comonotonic additivity of Tail-Value-at-Risk. Value-at-Risk enjoys neither, and the worst-case Value-at-Risk of a sum over the Fréchet class is in general strictly larger than the comonotonic value and is not attained by comonotonicity. Computing it is a genuine optimal transport problem, solved in practice by the rearrangement algorithm of Embrechts, Puccetti and Rüschendorf (12, 24), which iteratively rearranges the marginal samples to make the column sums as equal as possible. This is a central tool for quantifying model uncertainty in aggregate Value-at-Risk under the Solvency II and Basel frameworks.

## 6.5 Combining Models by Wasserstein Barycenters

Actuaries frequently need to combine several loss models, for instance the opinions of several experts, or mortality distributions from several data sources. Averaging densities pointwise produces a mixture, which is over-dispersed and typically leaves the parametric family. The Wasserstein barycenter provides a more faithful average that interpolates the distributions themselves.

**Definition 6.5** (Wasserstein barycenter). *Given loss models $\mu_1, \ldots, \mu_K$ and weights $w_k \geq 0$ with $\sum_k w_k = 1$, a Wasserstein barycenter is a minimiser*

$$\overline{\mu} \in \arg\min_{\nu} \sum_{k=1}^{K} w_k \, W_2(\nu, \mu_k)^2. \tag{66}$$

**Theorem 6.2** (One-dimensional barycenters are quantile averages). *For loss models on $\mathbb{R}$, the Wasserstein-2 barycenter is unique and its quantile function is the weighted average of the quantile functions,*

$$F_{\overline{\mu}}^{-1}(u) = \sum_{k=1}^{K} w_k \, F_{\mu_k}^{-1}(u), \qquad u \in (0,1). \tag{67}$$

*Proof.* By the one-dimensional formula $W_2(\nu, \mu_k)^2 = \int_0^1 |F_\nu^{-1}(u) - F_{\mu_k}^{-1}(u)|^2 \, du$, the objective is $\int_0^1 \sum_k w_k |F_\nu^{-1}(u) - F_{\mu_k}^{-1}(u)|^2 \, du$. For each fixed $u$ the integrand is minimised over the value $F_\nu^{-1}(u)$ by the weighted mean $\sum_k w_k F_{\mu_k}^{-1}(u)$, which is non-decreasing in $u$ and hence is a valid quantile function (1). □

**Example 6.8** (Blending two exponential models). Combine $\mu_1 =$ Exp with mean 1000 and $\mu_2 =$ Exp with mean 1200 with equal weights. By Theorem 6.2 the barycenter has quantile function $\frac{1}{2}(-1000 \ln(1-u)) + \frac{1}{2}(-1200 \ln(1-u)) = -1100 \ln(1-u)$, which is the quantile function of Exp with mean 1100. The Wasserstein barycenter of two exponential models is therefore again exponential, with the averaged mean, whereas the naive mixture $\frac{1}{2}\mu_1 + \frac{1}{2}\mu_2$ is a bimodal-tailed distribution that is no longer exponential. This is why barycentric averaging is preferable for combining expert loss models or mortality tables: it preserves the shape of the risk.

## 6.6 Convex Order, Stop-Loss Reinsurance and Martingale Transport

A final connection ties optimal transport to the classical actuarial theory of ordering risks. Recall that $X$ precedes $Y$ in convex order, written $X \leq_{cx} Y$, if $\mathbb{E}[\phi(X)] \leq \mathbb{E}[\phi(Y)]$ for every convex $\phi$, this expresses that $Y$ is more variable than $X$ while having the same mean. Convex order is the language of stop-loss comparisons in reinsurance.

**Definition 6.6** (Stop-loss transform). *The stop-loss transform of a loss $X$ at retention $d$ is $\pi_X(d) = \mathbb{E}[(X-d)_+]$, the expected cost to a reinsurer of a treaty paying the excess of $X$ over $d$. One has $X \leq_{cx} Y$ if and only if $\mathbb{E}[X] = \mathbb{E}[Y]$ and $\pi_X(d) \leq \pi_Y(d)$ for all $d$.*

**Example 6.9.** If $X$ is the fixed loss equal to its mean $\mathbb{E}[Y]$ almost surely and $Y$ is any risk with that mean, then $X \leq_{cx} Y$, since a degenerate risk is the least variable one with a given mean. In reinsurance terms, replacing a random loss by its mean can only reduce every stop-loss premium, which is the intuition that diversification and pooling reduce variability.

The link to transport is through Strassen's theorem: $X \leq_{cx} Y$ holds if and only if there is a martingale coupling of $X$ and $Y$, that is, a coupling $\pi \in \Pi(F_X, F_Y)$ under which $\mathbb{E}[Y \mid X] = X$. Optimising a transport cost over such martingale couplings is martingale optimal transport, the tool behind model-independent price and reserve bounds when only the marginals are trusted (24).

**Example 6.10.** Let $X = \delta_1$ be a unit loss with certainty and $Y = \frac{1}{2}\delta_0 + \frac{1}{2}\delta_2$. The two have the same mean 1, and the coupling that sends the mass at 1 equally to 0 and 2 is a martingale coupling, since $\mathbb{E}[Y \mid X = 1] = 1$. Hence $X \leq_{cx} Y$: the two-point risk is a mean-preserving spread of the certain loss, and any convex reinsurance cost is larger for $Y$ than for $X$.

### 6.7 A Note on the Non-Commutative Case

It is natural to ask whether the non-commutative optimal transport of Section 5 has actuarial applications. At present the connection is speculative. Quantum probabilistic models of insurance dependence are not part of standard actuarial practice, and the quantum Wasserstein distances were developed with quantum information theory rather than risk management in mind. What can be said with confidence is structural: the coupling formulation of dependence, the Kantorovich duality behind robust bounds, and the barycenter construction for model combination are all instances of the general operator-algebraic framework, of which the classical actuarial applications above are the commutative case. Whether the genuinely non-commutative features, namely the off-diagonal correlations with no classical analogue, carry actuarial meaning is an open question that we flag as speculative rather than established.

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